



Thermodynamics and Thermoelasticity beyond the Harmonic Approximation

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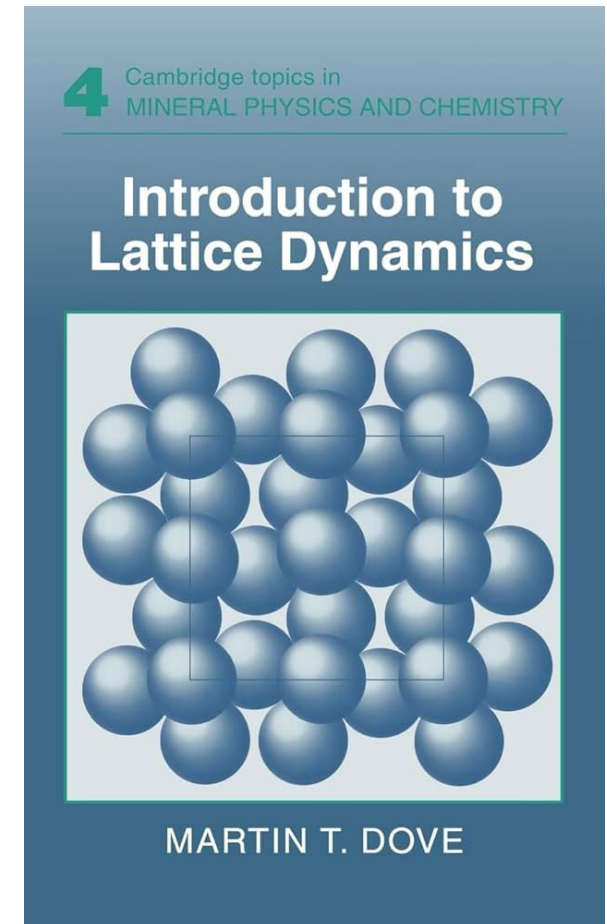
Temperature and Lattice Dynamics



An effective inclusion of the effect of **temperature** requires a description of the motion of the atomic nuclei within the solid (i.e. **lattice dynamics**).

Indeed, atoms in a lattice are never at rest in their equilibrium positions: they always move of vibrational motion (even at 0 K because of zero-point motion). Such lattice vibrations (**phonons**) are key to understand:

- Thermodynamics (specific heat, free energy, entropy)
- Phase transitions
- Thermal expansion and Thermo-elasticity
- Lattice thermal conductivity and Pyro-electricity
- Debye-Waller factors and ADPs
- Thermal broadening of spectral peaks
- and many other properties of solids



Vibrations in Solids: Phonon Dispersion

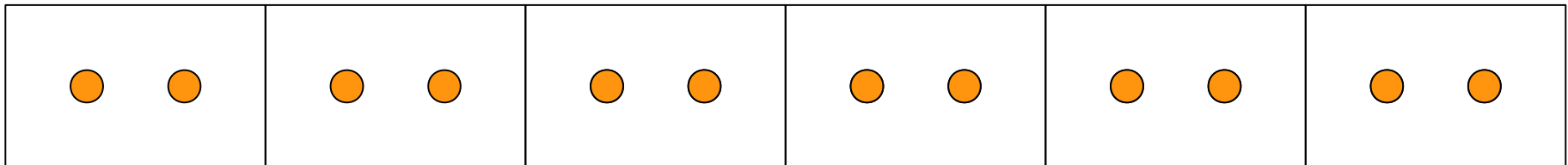


In a molecule, there is a finite number of vibrational degrees of freedom (i.e. distinct ways in which the atoms can vibrate relative to one another), and we know how to count them: it is $N_{\text{vib}} = 3N - 6(5)$ for a molecule with N atoms.



For instance, for a linear molecule with $N = 2$, we have $N_{\text{vib}} = 1$, i.e. just the stretching mode, vibrating at frequency ω .

In an infinite periodic lattice, the number of vibrational degrees of freedom becomes infinite, $N_{\text{vib}} = \infty$. Indeed atoms in different lattice cells can vibrate with phase modulations, leading to phonon dispersion.



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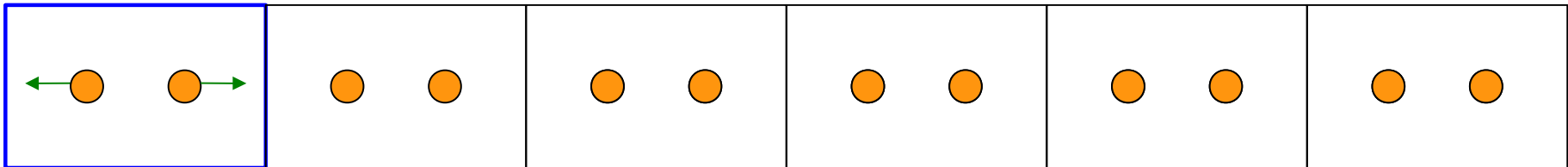


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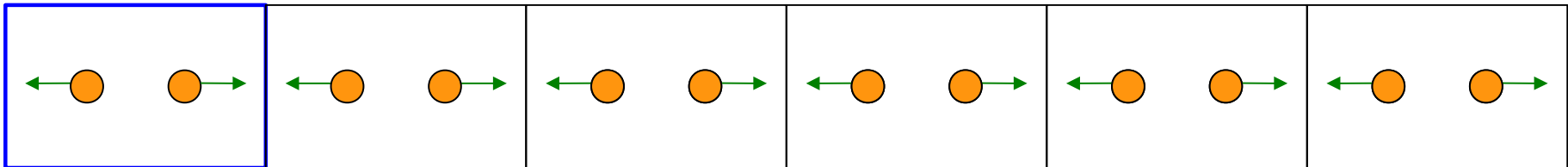


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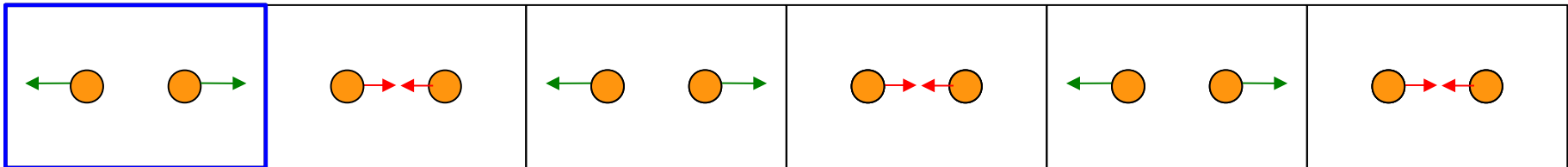


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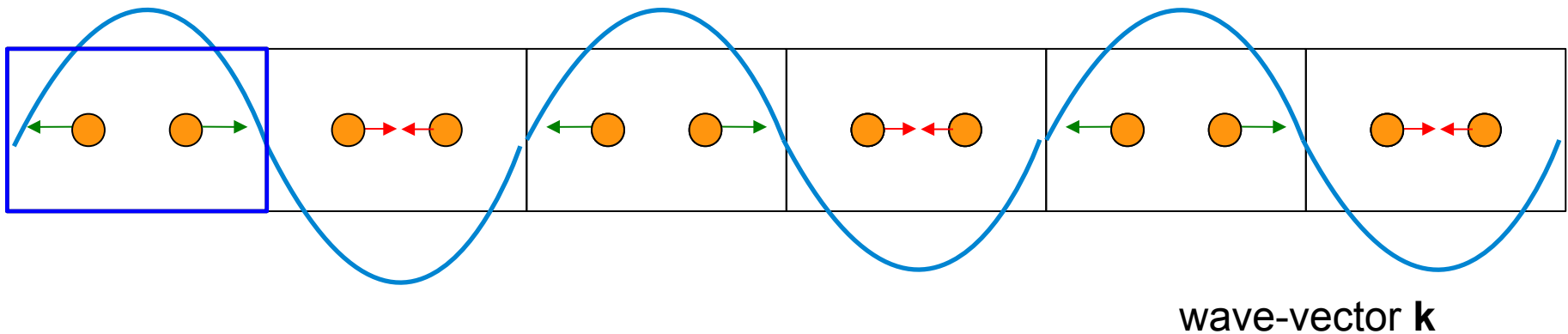


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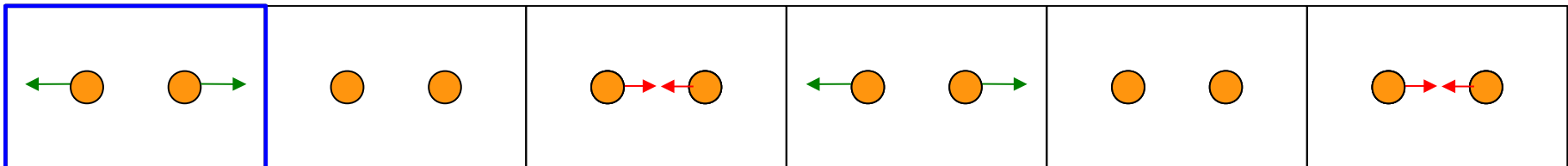


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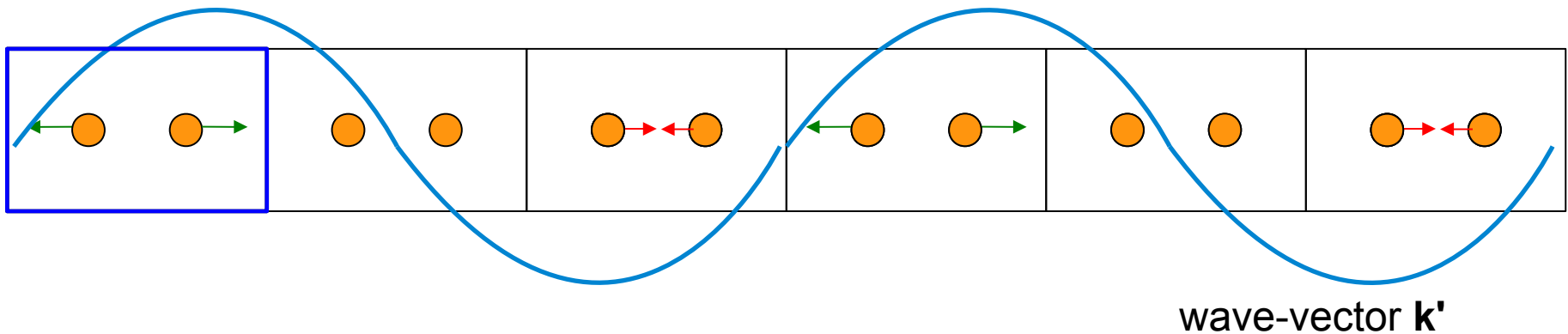


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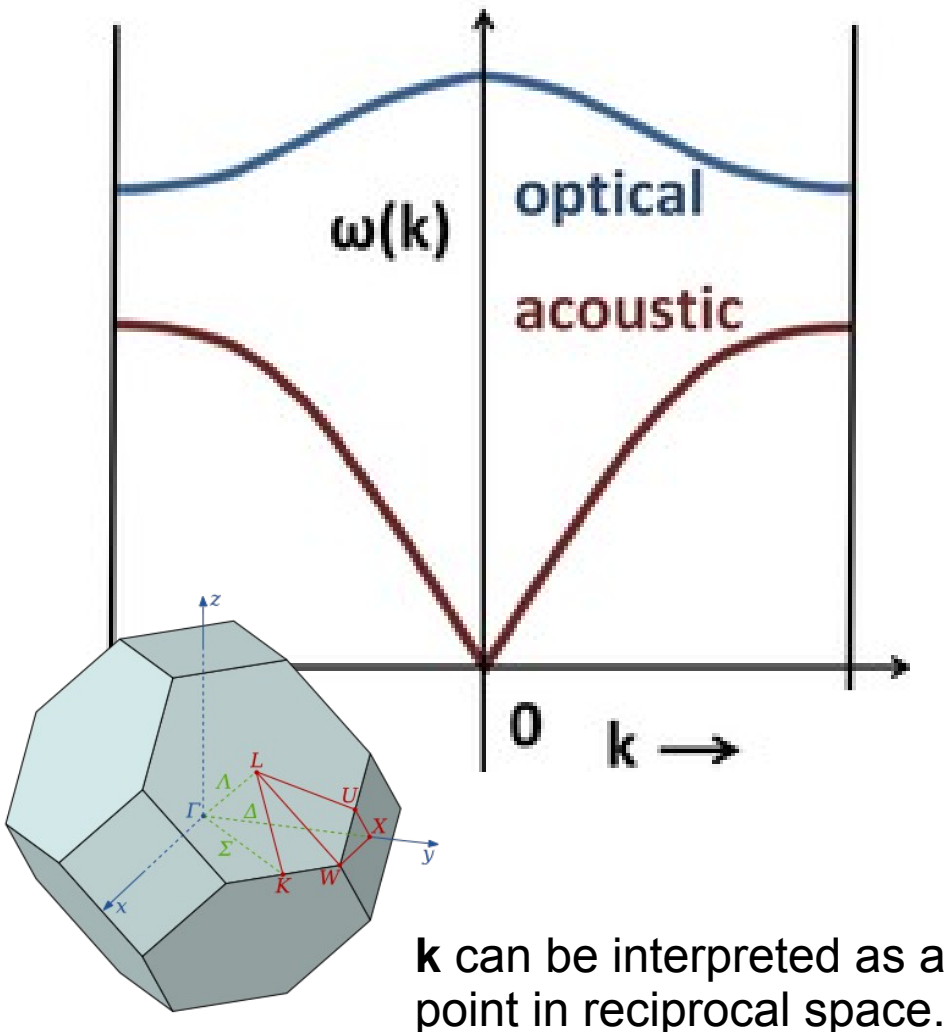
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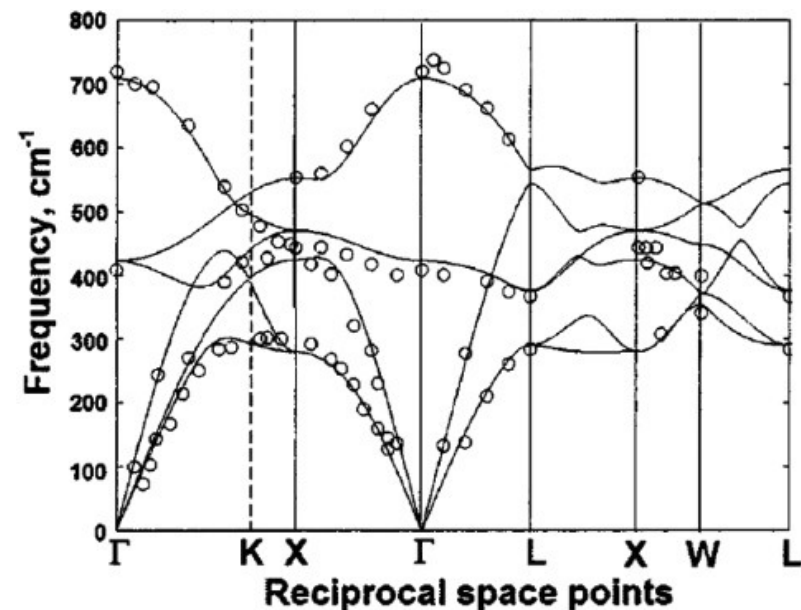


Thus, in a solid, vibration frequencies depend on wavevector $\mathbf{k}$, with a typical dispersion pattern, the so-called **phonon dispersion** $\omega(\mathbf{k})$.



Lattice vibrations at $\mathbf{k} = \mathbf{0}$ (i.e. at Γ) are probed by infrared (IR) and Raman spectroscopies.

Phonon dispersion can be probed by inelastic neutron scattering (INS).





Usually, nuclear motion is described within the **Born-Oppenheimer approximation** (i.e. decoupled from electronic motion). The electronic Schroedinger equation is solved at fixed nuclear configurations:

$$H^{\text{ele}}(\mathbf{r}; \mathbf{R})\Psi(\mathbf{r}; \mathbf{R}) = E(\mathbf{r}; \mathbf{R})\Psi(\mathbf{r}; \mathbf{R})$$

where $\mathbf{r}$ are electronic coordinates and $\mathbf{R}$ are nuclear coordinates. Then the nuclear Schroedinger equation is solved:

$$H^{\text{nuc}}(\mathbf{R})\Psi(\mathbf{R}) = \mathcal{E}(\mathbf{R})\Psi(\mathbf{R})$$

where the nuclear Hamiltonian is defined via the PES as:

$$H^{\text{nuc}}(\mathbf{R}) = T^{\text{nuc}}(\mathbf{R}) + E(\mathbf{R})$$



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Computing Lattice Dynamics: Harmonic App.



The nuclear Schroedinger equation is commonly solved by resorting to the **harmonic approximation**, i.e. by expanding the PES in a Taylor's series in the atomic Cartesian displacements $\mathbf{u}$, and truncating to second order:

$$E(\mathbf{R}) \simeq \sum_{i,j} \left(\frac{\partial^2 E}{\partial u_i \partial u_j} \right) \mathbf{u}_i \mathbf{u}_j$$

In this way, the nuclear dynamics is described in terms of a set of independent quantum harmonic oscillators, with exact known solutions for energy and wave-functions.

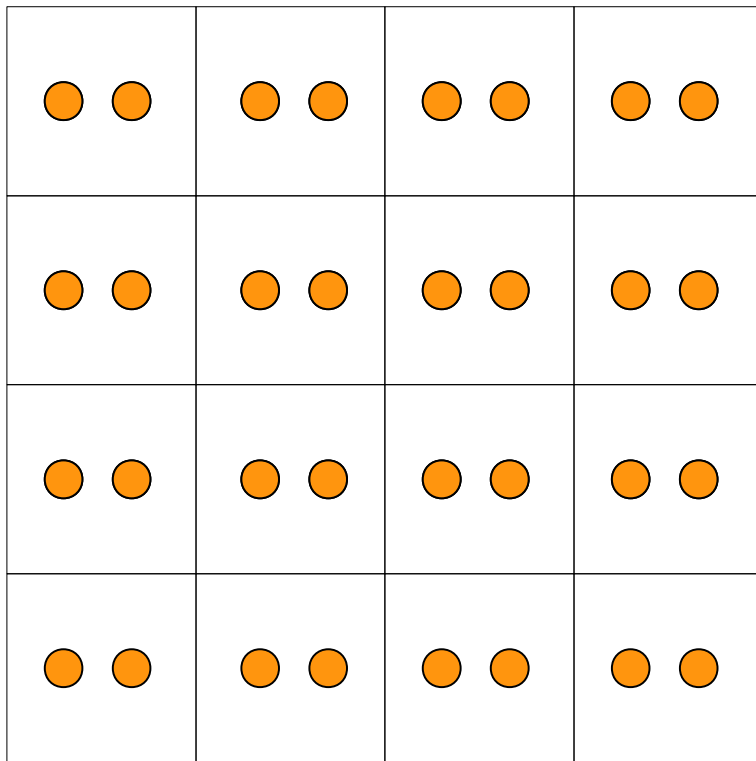
For a finite molecular system, harmonic vibration frequencies ω_i and normal modes are obtained from the diagonalization of the mass-weighted Hessian matrix of second energy derivatives with respect to pairs of atomic displacements:

$$H_{ij} = \frac{\partial^2 E}{\partial u_i \partial u_j} \quad \mathbf{W} \rightarrow W_{ij} = \frac{H_{ij}}{\sqrt{M_i M_j}} \quad \text{then} \quad \mathbf{WU} = \mathbf{\Lambda U}$$

Direct Space Approach to Phonon Dispersion



In CRYSTAL, a direct space (**frozen phonon**) approach is implemented where the interatomic force constants are first computed in direct space (in general within a **supercell**) and then Fourier transformed to reciprocal space where the dynamical matrices are built and diagonalized:



$$H_{ij}^{\mathbf{g}} = \frac{\partial^2 E}{\partial u_{i0} \partial u_{j\mathbf{g}}}$$



$$\mathbf{W}^{\mathbf{k}} \rightarrow W_{ij}^{\mathbf{k}} = \sum_{\mathbf{g}} \frac{H_{ij}^{\mathbf{g}}}{\sqrt{M_i M_j}} e^{i\mathbf{k} \cdot \mathbf{g}}$$

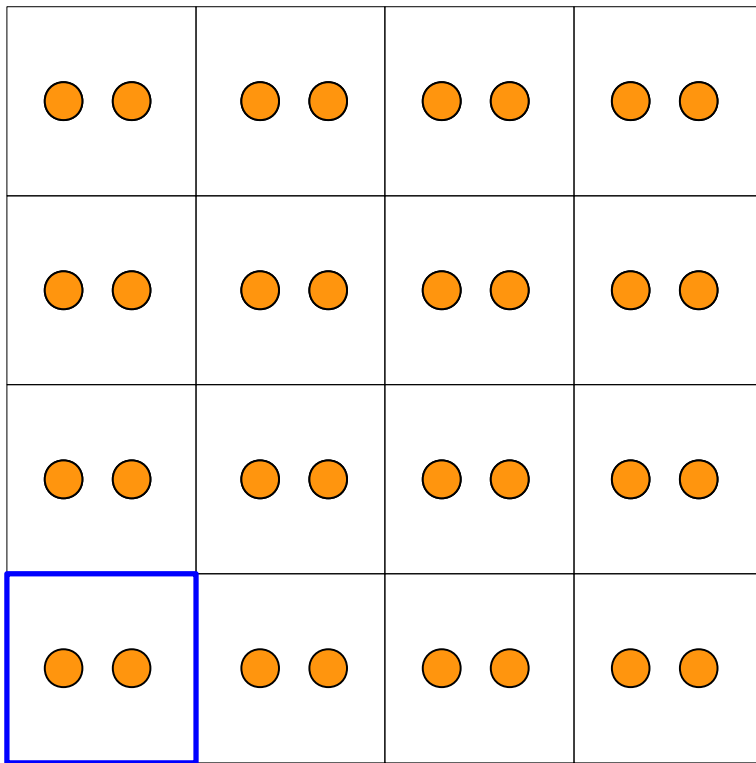


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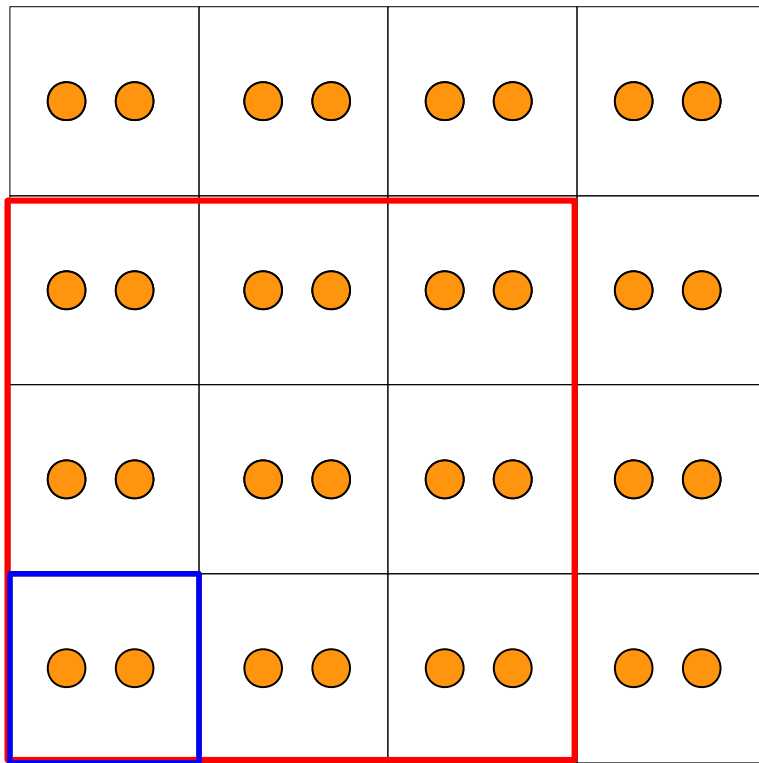


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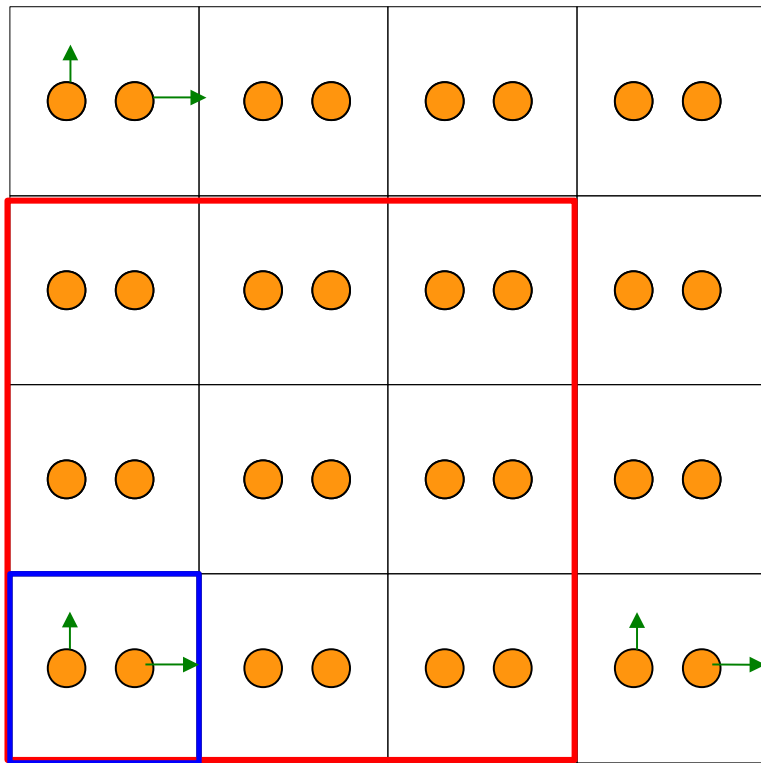


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Atoms are displaced only in the blue cell.

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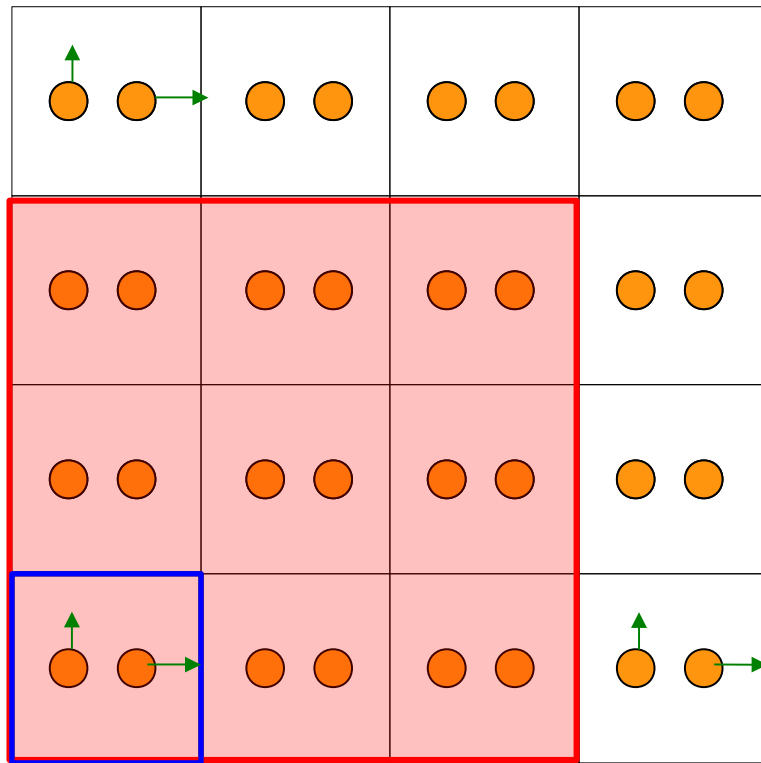


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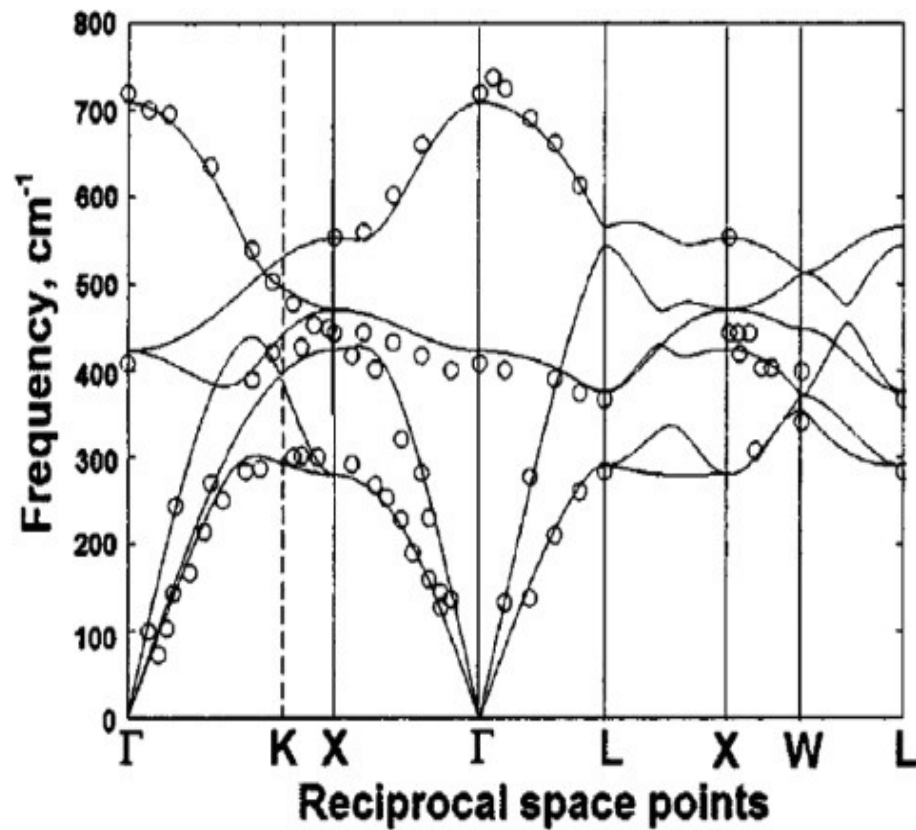
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The forces acting on all the atoms in the red cell are computed.

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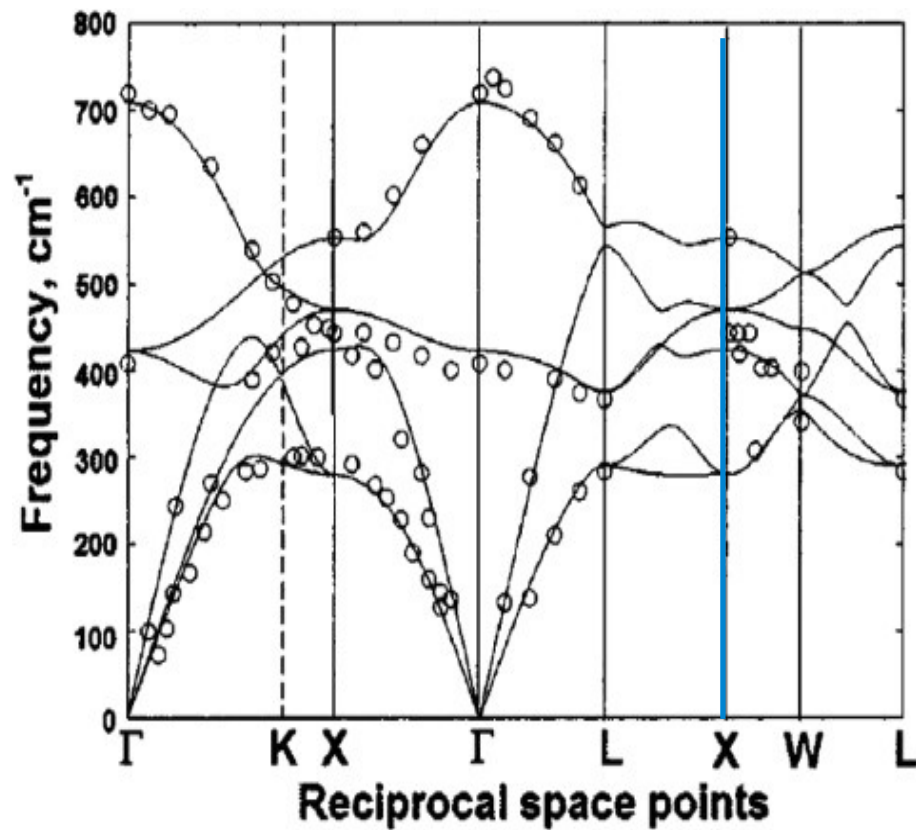
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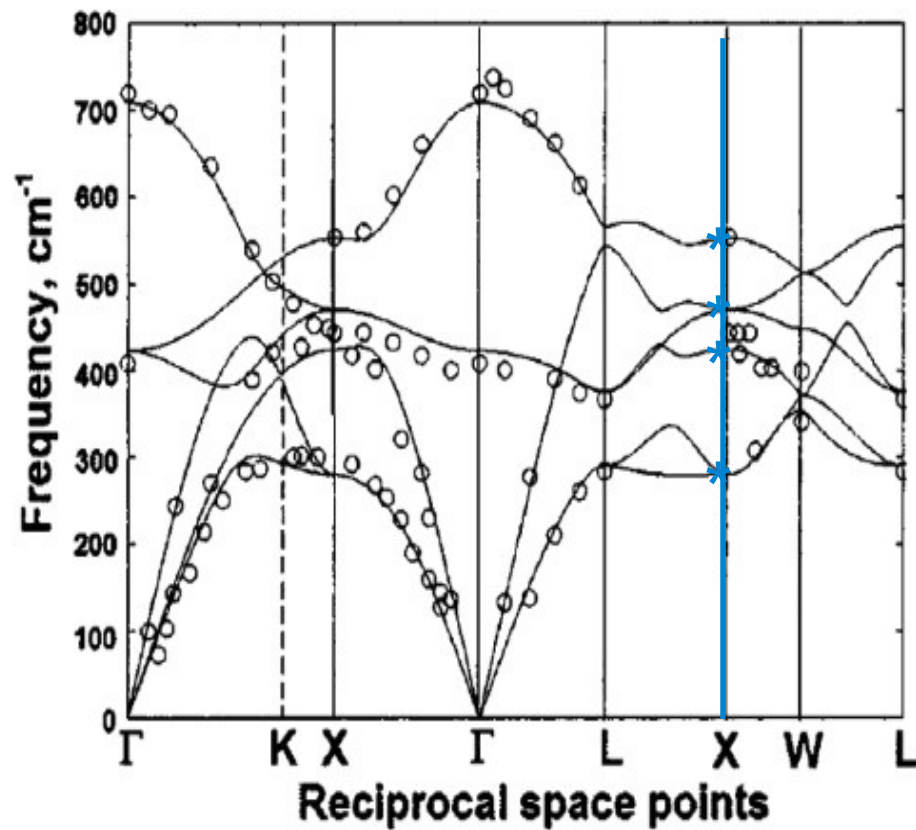
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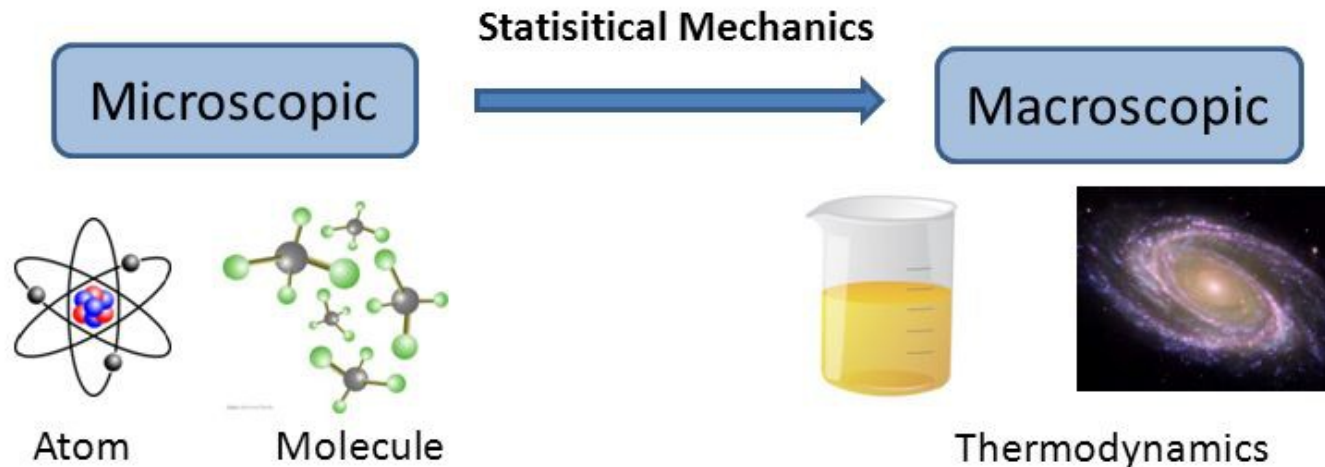
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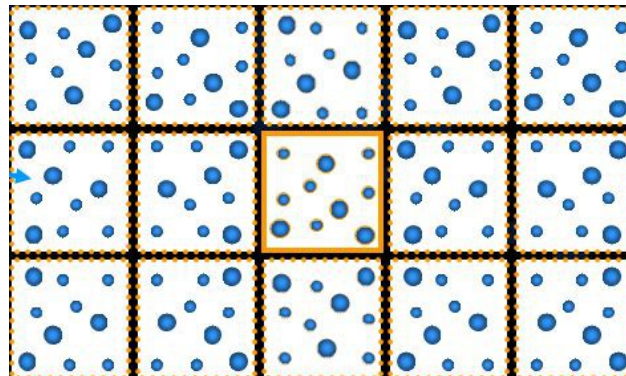
From Phonons to Thermodynamics



Statistical thermodynamics provides a link between a quantum-mechanical description of the microscopic states and the thermodynamic properties of a macroscopic system.



From the **ergodic hypothesis**, macroscopic thermodynamic properties can be obtained as averages over **ensembles**.



From Phonons to Thermodynamics



The **partition function** of the **canonical** ensemble associated to harmonic vibrational degrees of freedom of a solid is defined as follows from phonon frequencies:

$$Z_{\text{vib}}(T) = \prod_{\mathbf{k}} \prod_i \sum_{n=0}^{\infty} e^{-\frac{(\frac{1}{2} + n)\hbar\omega_{\mathbf{k}i}}{k_B T}}$$

All thermodynamic state functions can be derived from it. For instance:

$$U_{\text{vib}}(T) = -\frac{d \ln Z_{\text{vib}}}{d\beta}$$

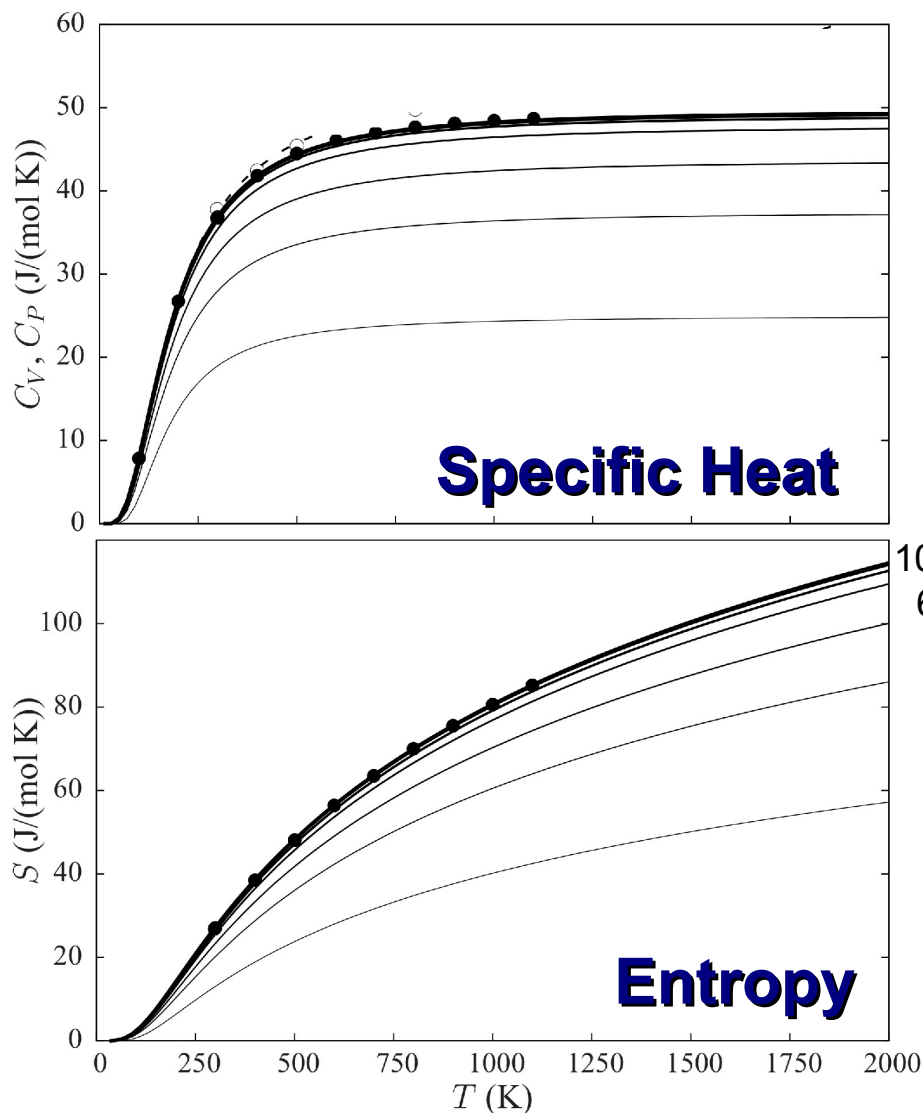
$$S_{\text{vib}}(T) = \frac{U_{\text{vib}}}{T} + k_B \ln Z_{\text{vib}}$$

$$F_{\text{vib}}(T) = -k_B T \ln Z_{\text{vib}}$$

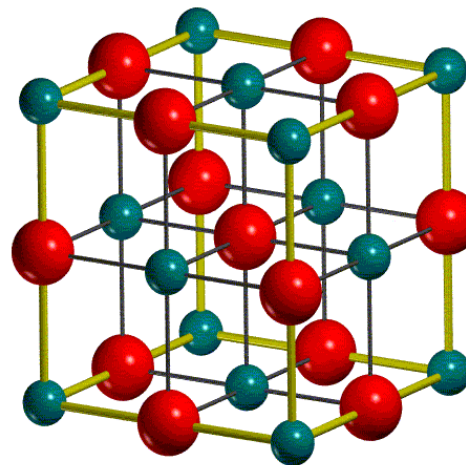
Harmonic Thermodynamics



How large should the supercell be?



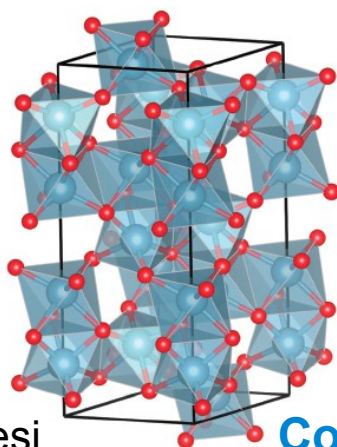
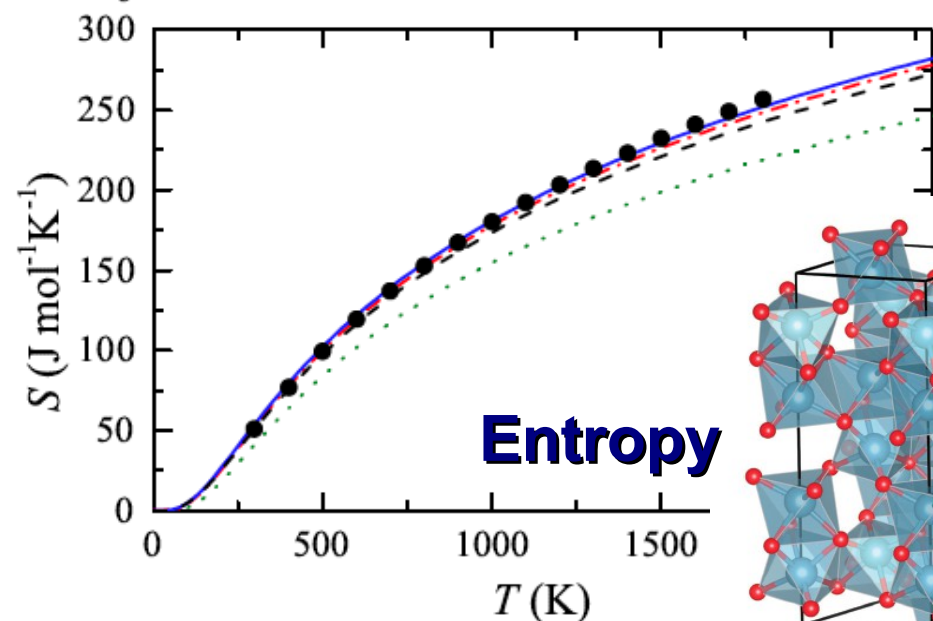
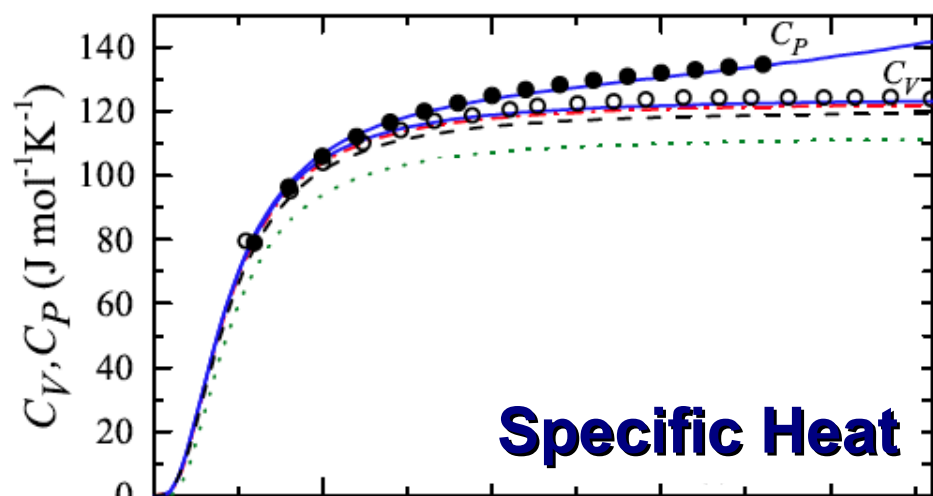
108 k points / 216 atoms
64 k points / 128 atoms
8 k points / 16 atoms
4 k points / 8 atoms
1 k point / 2 atoms



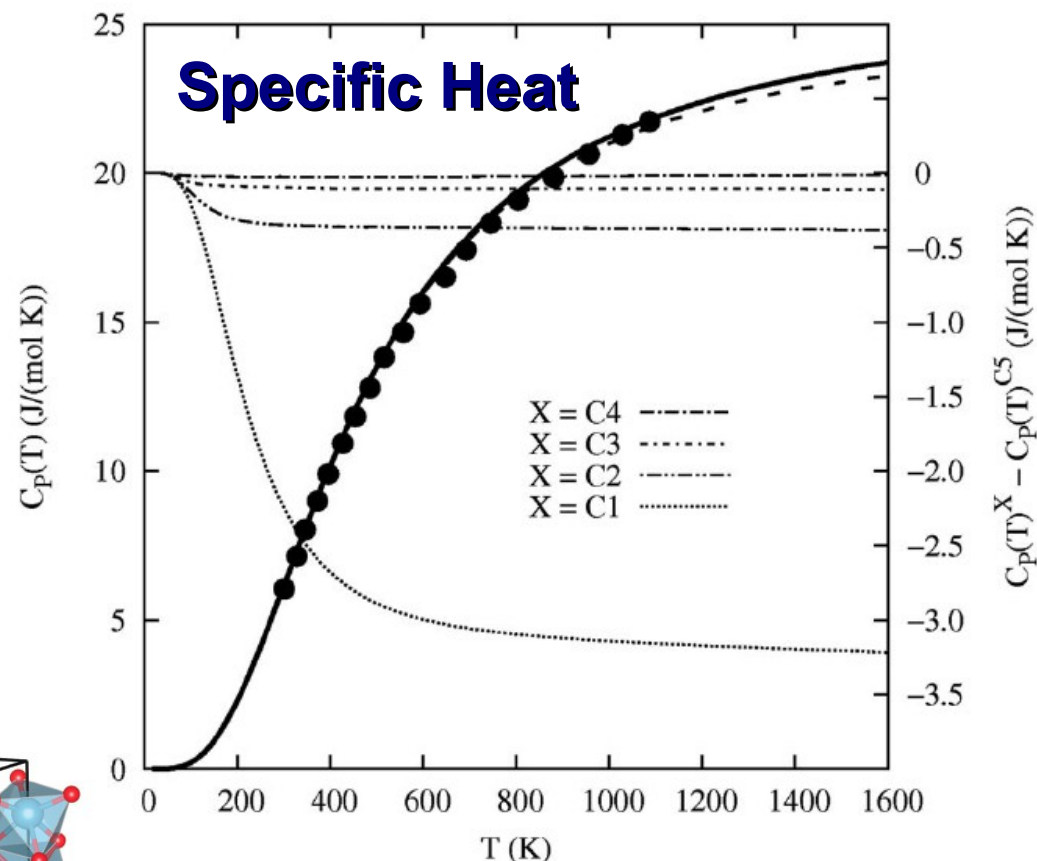
MgO

AE, M. Shahrokhi, R. Moradian and R. Dovesi,
J. Chem. Phys., **142**, 044114 (2015)

Harmonic Thermodynamics



Corundum



AE, *J. Chem. Phys.*, **141**, 124115 (2014)

Diamond

AE, J. Maul, R. Demichelis and R. Dovesi, *Phys. Chem. Chem. Phys.*, **17**, 11670-11677 (2015)

Limits of the Harmonic Approximation



The harmonic approximation has two main limitations:

1) By neglecting higher-than-quadratic terms of the PES, **vibration modes are uncoupled**. As a result:

- Infinite lattice thermal conductivity
- Infinite phonon lifetime
- Zero pyroelectricity
- No mode couplings (i.e. no combination bands, overtones, resonances in spectra)
- ...

2) Its description of the lattice dynamics is at **constant volume**. As a result:

- Zero thermal expansion
- Elastic constants independent of temperature
- $C_p = C_v$
- Impossibility to combine temperature and pressure
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See next talk by Davide Mitoli

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The second class of limitations of the HA has mostly to do with **thermal and thermodynamic properties** and can be effectively overcome (at least within a certain temperature range) through the so-called **quasi-harmonic approximation** (QHA):

$$F_{\text{vib}}^{\text{HA}}(T) = U_{\text{zp}} + k_B T \sum_{\mathbf{k}, i} \ln \left[1 - e^{-\frac{\hbar \omega_{\mathbf{k}i}}{k_B T}} \right]$$

$$F_{\text{vib}}^{\text{QHA}}(T, V) = U_{\text{zp}}(V) + k_B T \sum_{\mathbf{k}, i} \ln \left[1 - e^{-\frac{\hbar \omega_{\mathbf{k}i}(V)}{k_B T}} \right]$$

The same harmonic expressions for all thermodynamic functions are retained but they gain an additional dependence on volume. The following can be computed:

- Thermal expansion
- P-V-T equation of state
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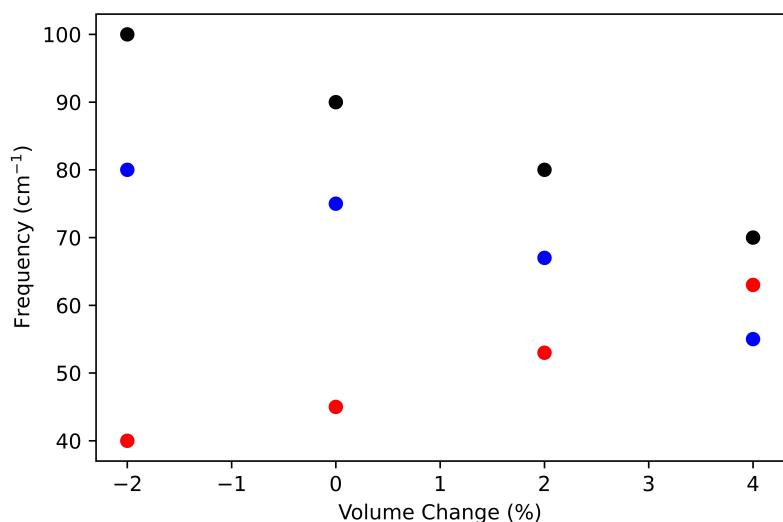
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Quasi-Harmonic Approximation



Harmonic phonons are computed at different volumes. In CRYSTAL, a fully automated algorithm has been implemented. Harmonic frequencies are computed at 4 volumes (expanded and compressed) and individually fitted.



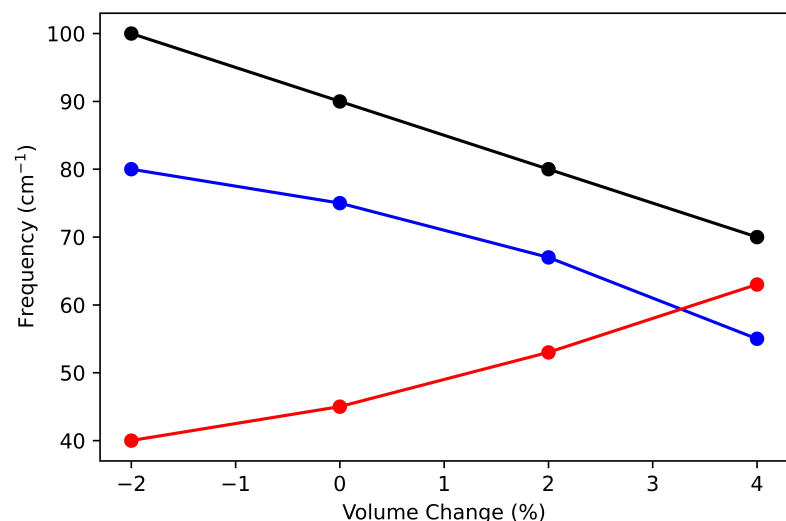
```
QHA
STEP
2
POINTS
4
END
```

AE, *J. Chem. Phys.*, **141**, 124115 (2014)

Quasi-Harmonic Approximation



Harmonic phonons are computed at different volumes. In CRYSTAL, a fully automated algorithm has been implemented. Harmonic frequencies are computed at 4 volumes (expanded and compressed) and individually fitted.



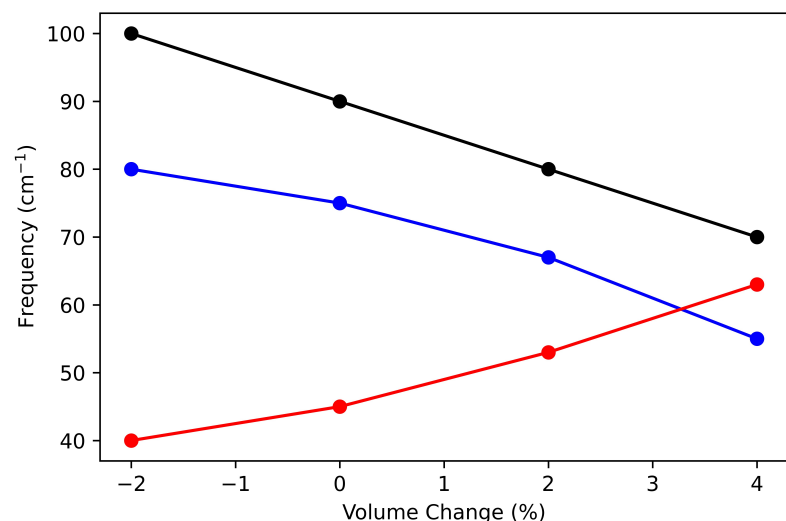
```
QHA  
STEP  
2  
POINTS  
4  
END
```

AE, *J. Chem. Phys.*, **141**, 124115 (2014)

Quasi-Harmonic Approximation



Harmonic phonons are computed at different volumes. In CRYSTAL, a fully automated algorithm has been implemented. Harmonic frequencies are computed at 4 volumes (expanded and compressed) and individually fitted.



Gruneisen parameter > 0

Gruneisen parameter > 0

Gruneisen parameter < 0

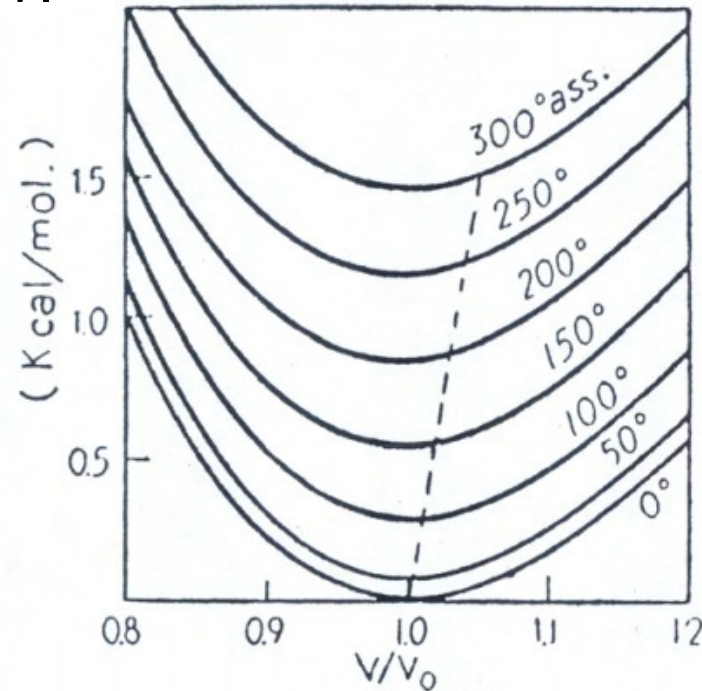
```
QHA
STEP
2
POINTS
4
END
```

AE, *J. Chem. Phys.*, **141**, 124115 (2014)

Thermal Expansion



The equilibrium volume $V(T)$ at a given temperature T is obtained by minimizing F with respect to V at each T .



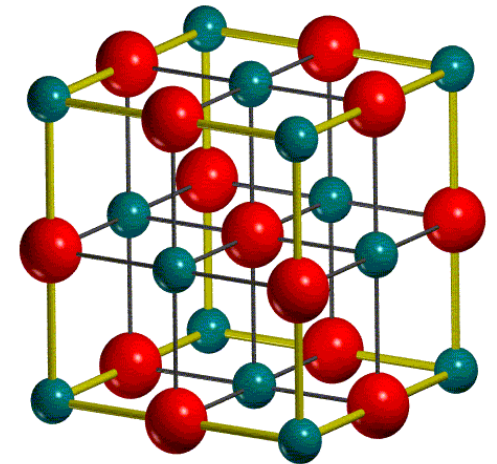
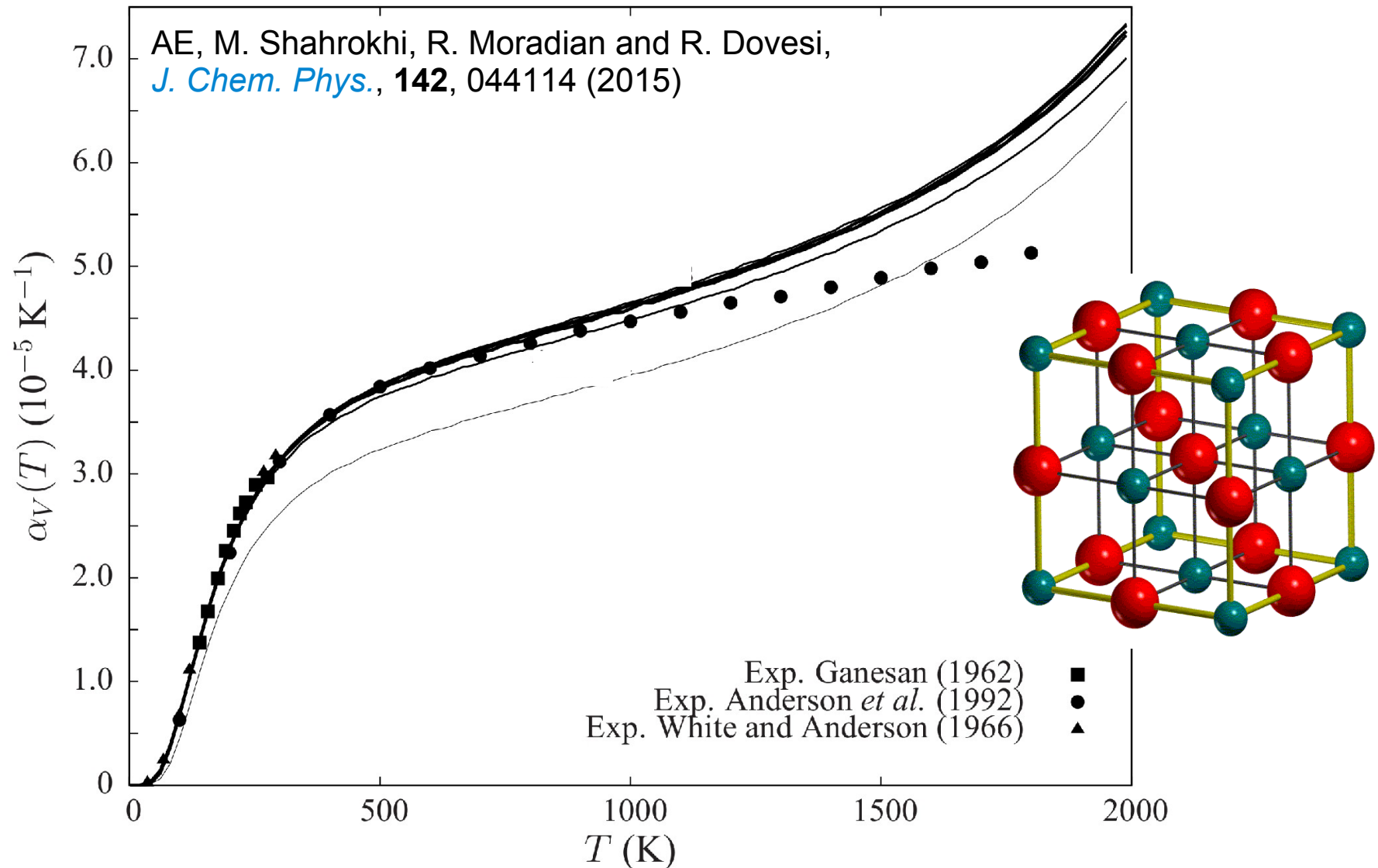
From the $V(T)$ relation, the **thermal expansion coefficient** is defined as:

$$\alpha_V(T) = \frac{1}{V(T)} \left(\frac{\partial V(T)}{\partial T} \right)_{P=0}$$

Validity Domain of QHA



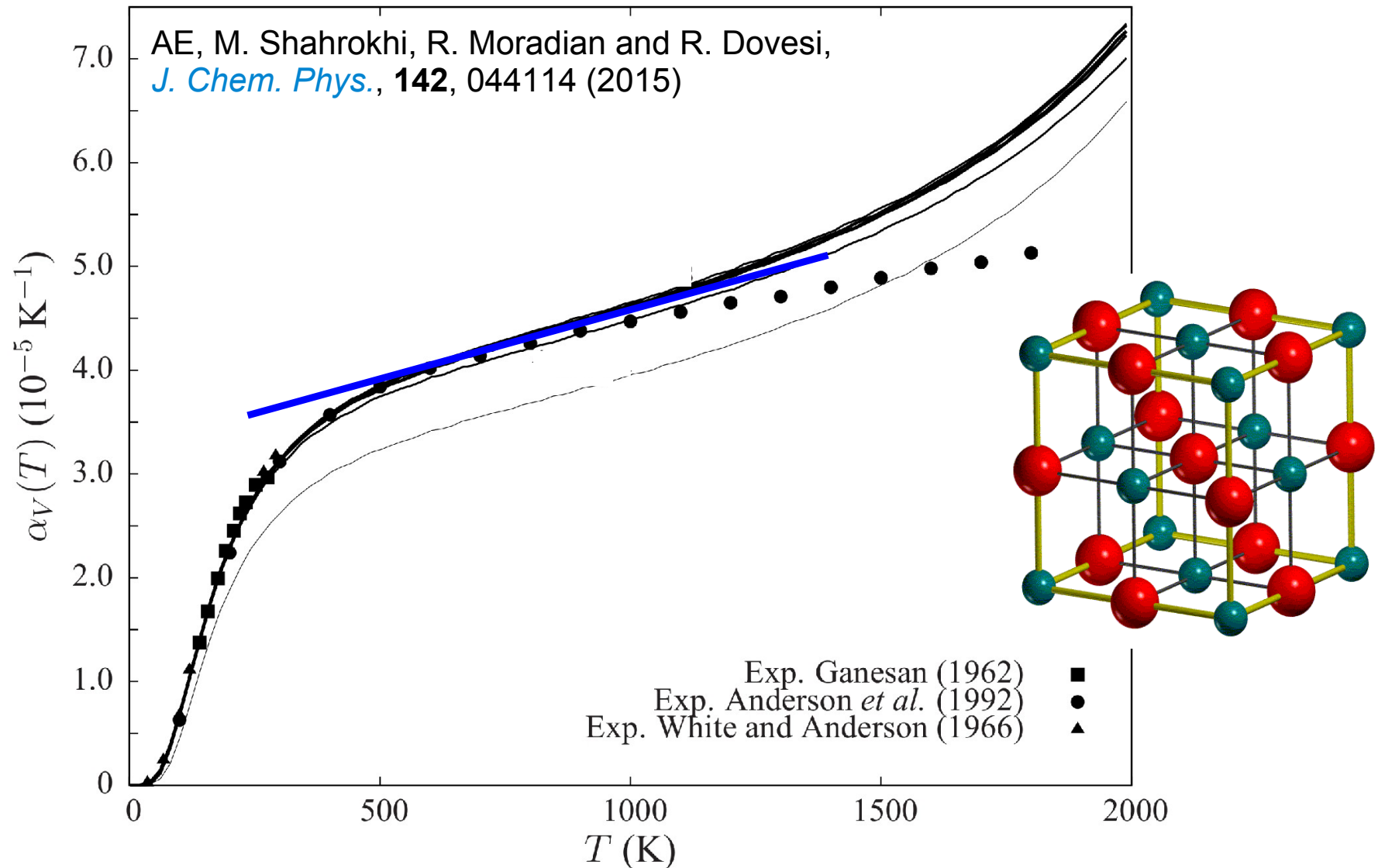
Is the QHA valid at any temperature? Generally not. It depends on the system and on pressure.



Validity Domain of QHA



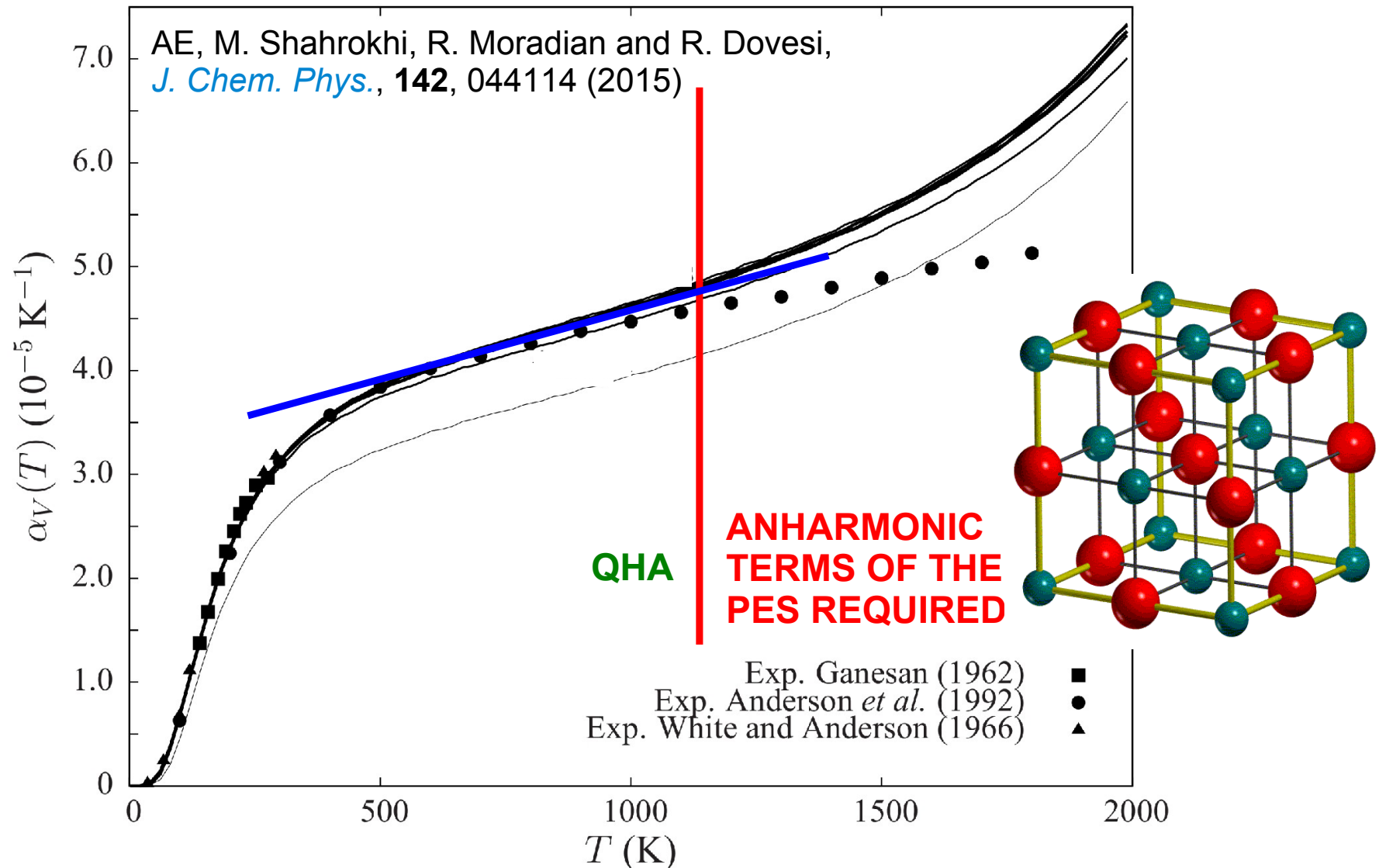
Is the QHA valid at any temperature? Generally not. It depends on the system and on pressure.



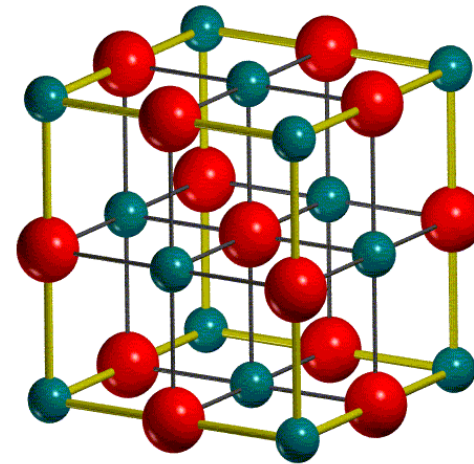
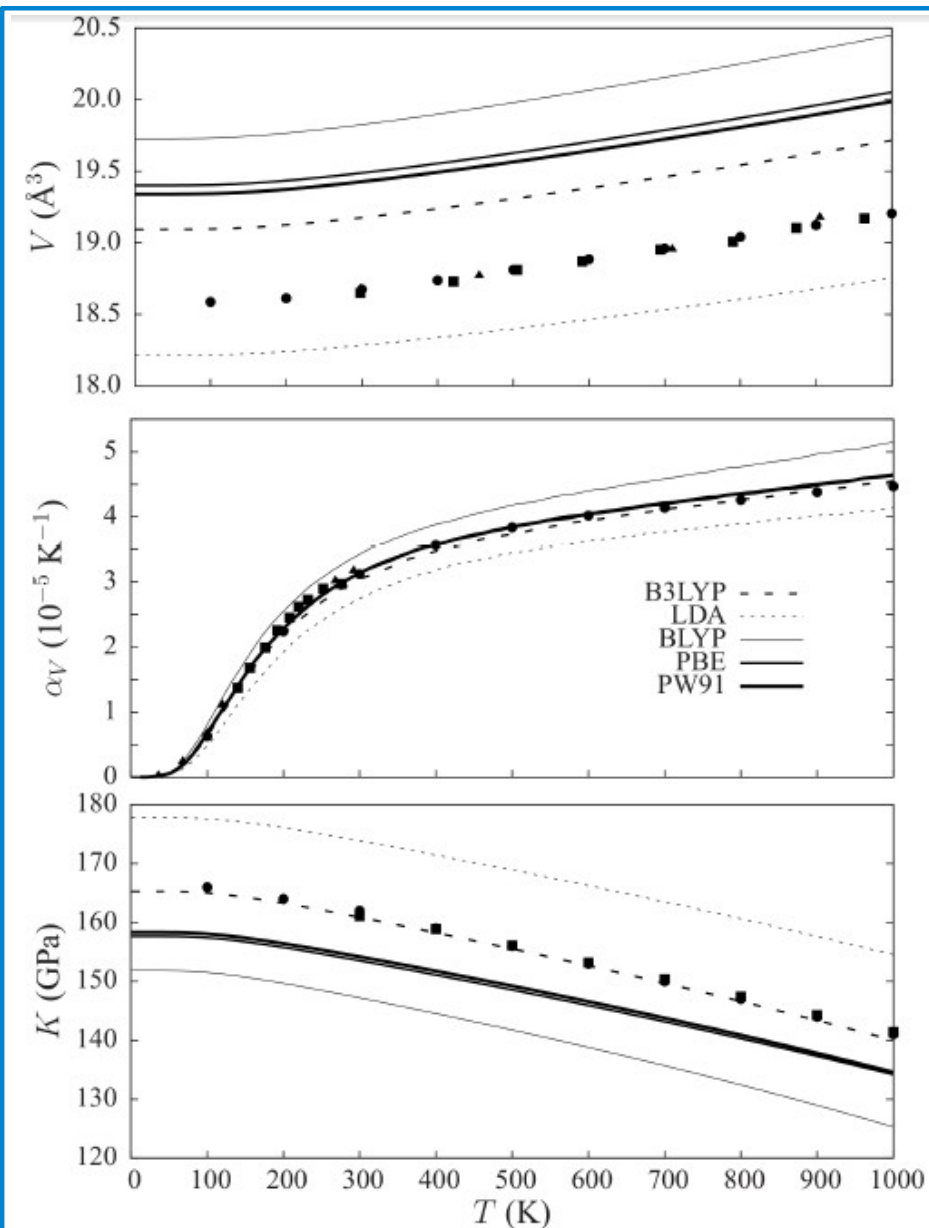
Validity Domain of QHA



Is the QHA valid at any temperature? Generally not. It depends on the system and on pressure.



Structural and Mechanical Properties



MgO

Bulk Modulus $K(T)$

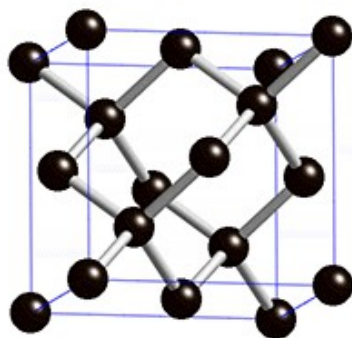
(a measure of the resistance of a substance to bulk compression)

$$K(T) = V(T) \left(\frac{\partial^2 F^{\text{QHA}}(V; T)}{\partial V^2} \right)_T$$

Structural and Mechanical Properties



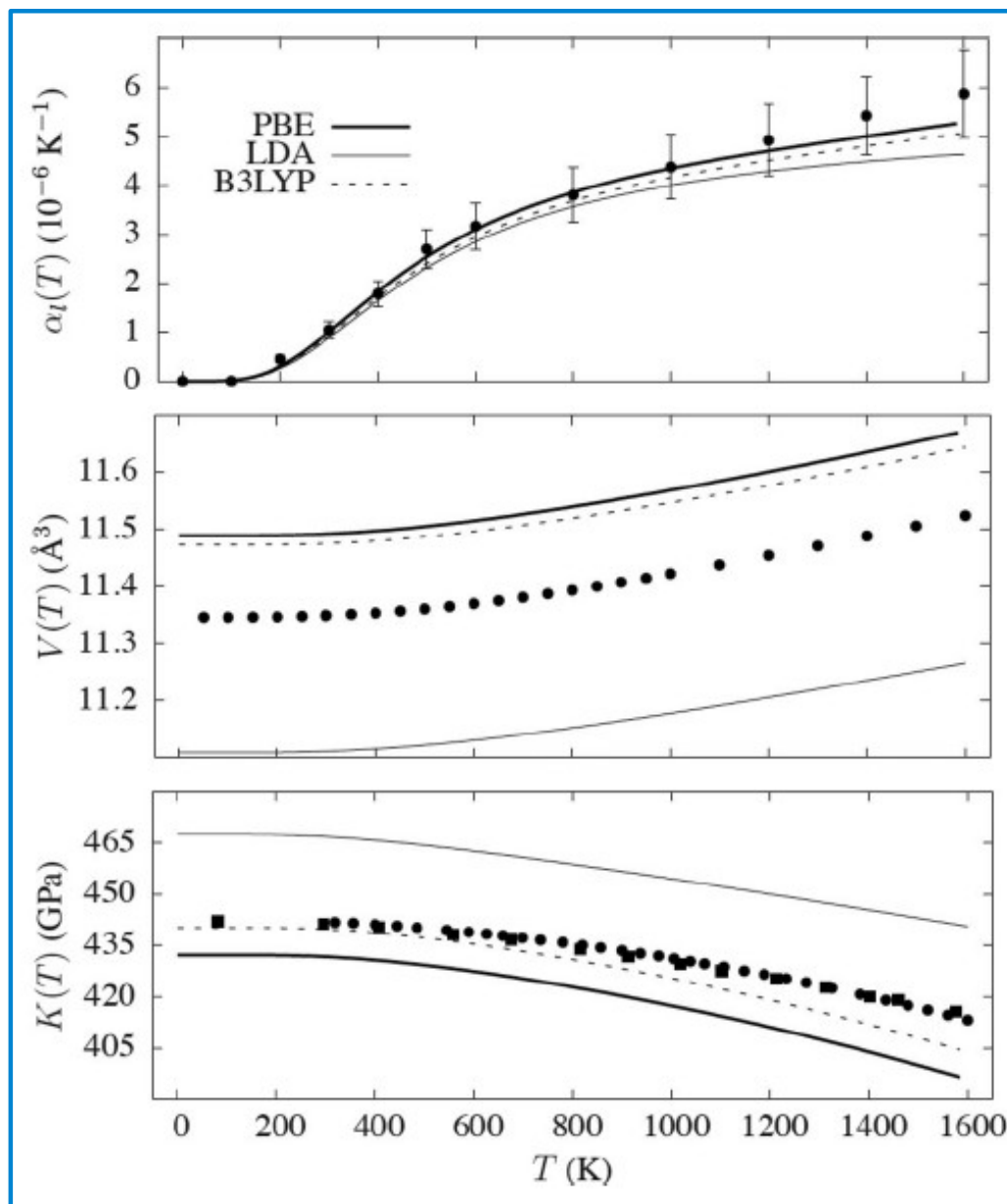
AE, *J. Chem. Phys.*, **141**, 124115 (2014)



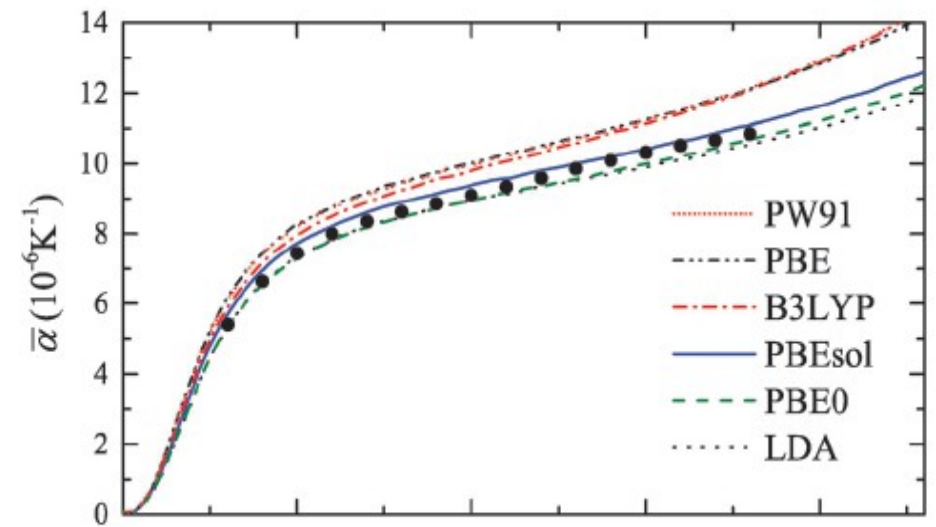
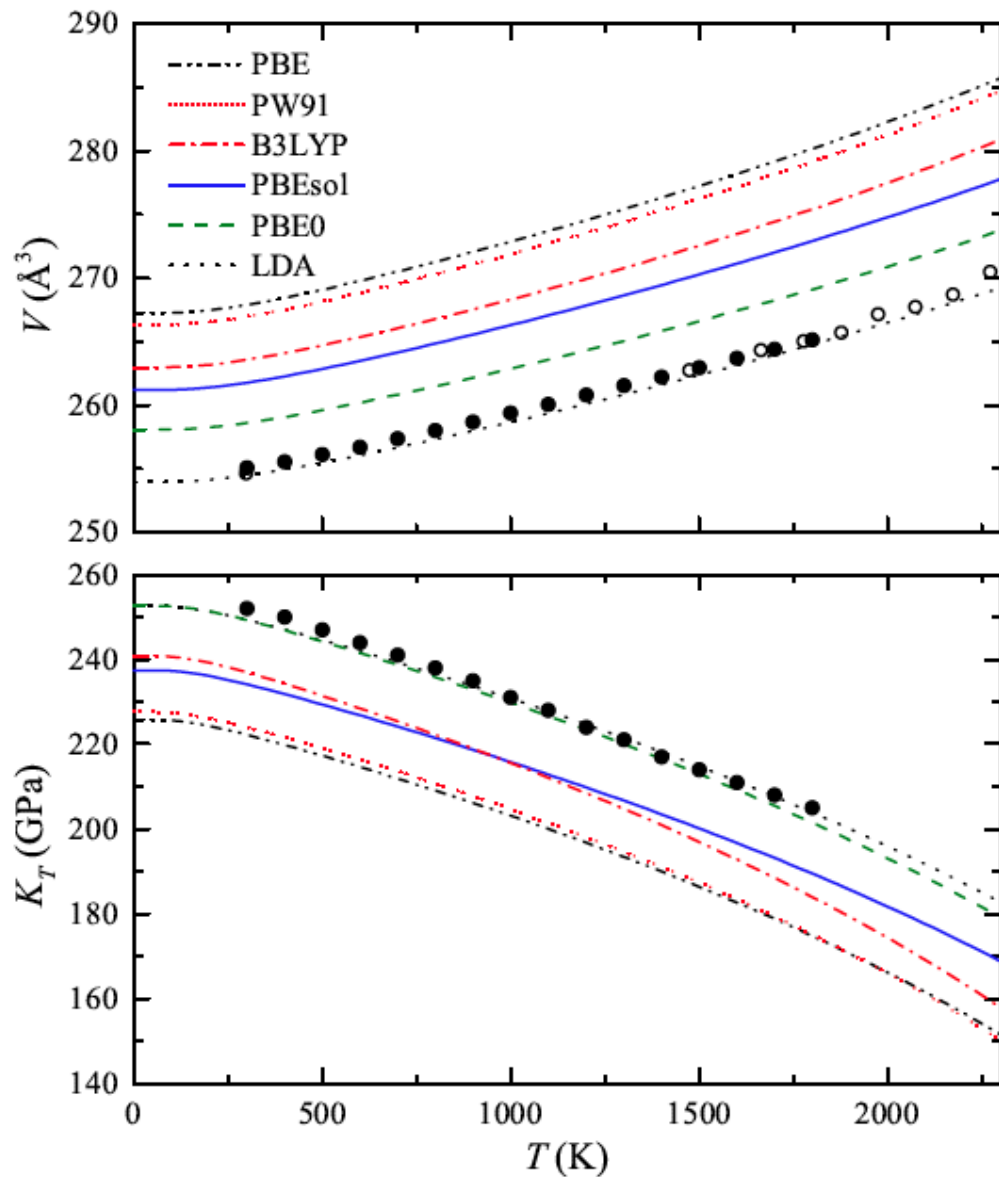
Diamond

TABLE I. Equilibrium volume of the primitive cell, V_0 (in $\text{\AA}^3$), and bulk modulus, K_0 (in GPa), of diamond at zero pressure and temperature, as computed with different Hamiltonians and compared with experimental values.^{50,76} Data in parentheses are obtained by neglecting ZPM effects. The lattice parameter is given by $a_0 = (4V_0)^{1/3}$.

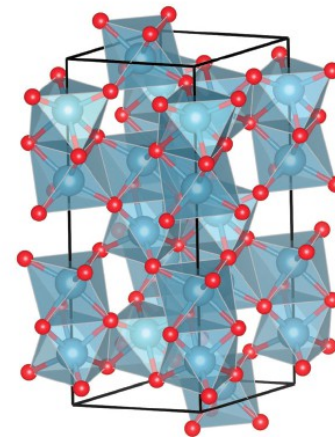
	LDA	PBE	B3LYP	Exp.
V_0	11.11 (10.99)	11.49 (11.36)	11.48 (11.35)	11.34
K_0	467.5 (476.8)	431.2 (441.5)	439.5 (449.2)	442.1



Structural and Mechanical Properties



AE, J. Maul, R. Demichelis and R. Dovesi,
Phys. Chem. Chem. Phys., **17**, 11670-11677 (2015)

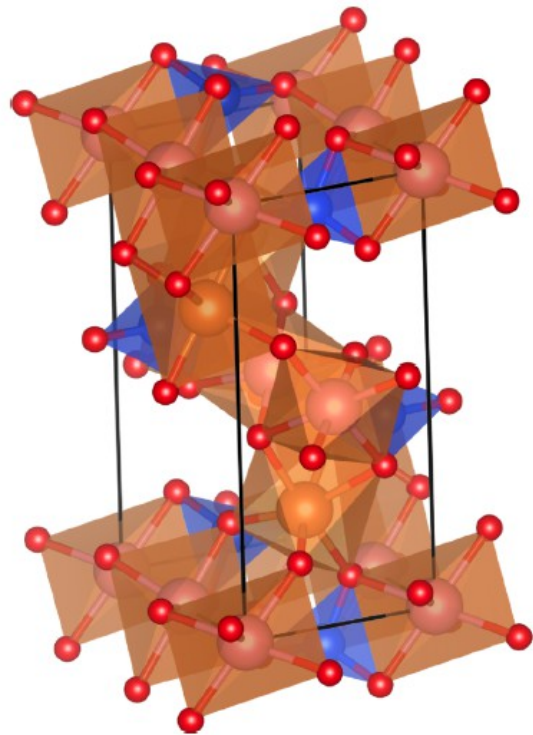


Corundum Al_2O_3

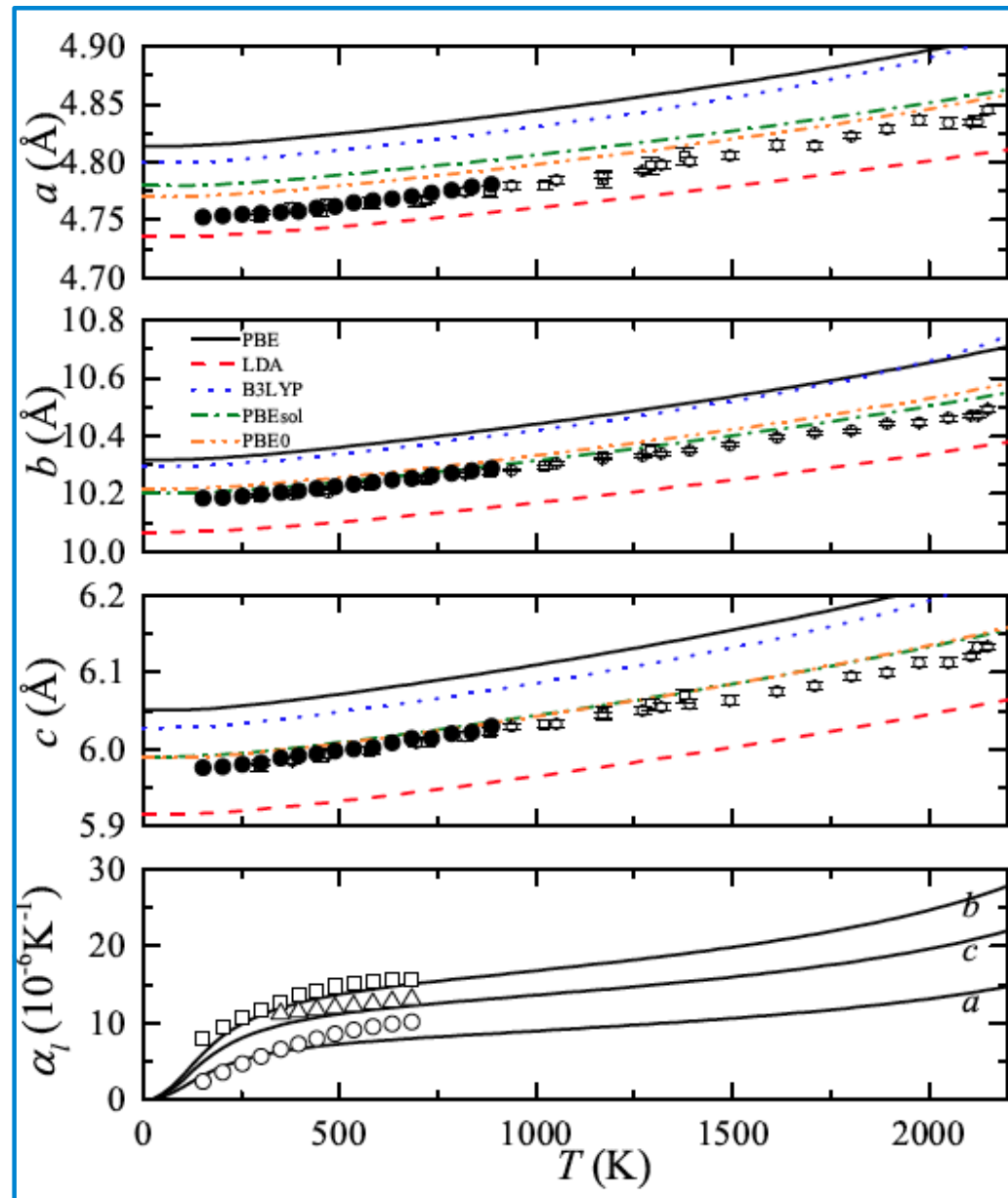
Anisotropic Thermal Expansion



AE, J. Maul, M. De la Pierre and R. Dovesi,
J. Chem. Phys., **142**, 204502 (2015)



Forsterite Mg_2SiO_4



Quasi-Harmonic Approximation



Thermal expansion

Molecular crystals still represent a challenge to state-of-the-art quantum-chemical methods:

- 1) A **balanced description** of many **chemical interactions** (including weak ones) is needed. We use Grimme's **DFT-D3**.
- 2) They are typically characterized by **soft phonon modes** (high numerical precision needed).
- 3) **Low symmetry**, **more atoms** than simple inorganic solids.
- 4) **High anharmonicity** of optical modes (many hydrogens).

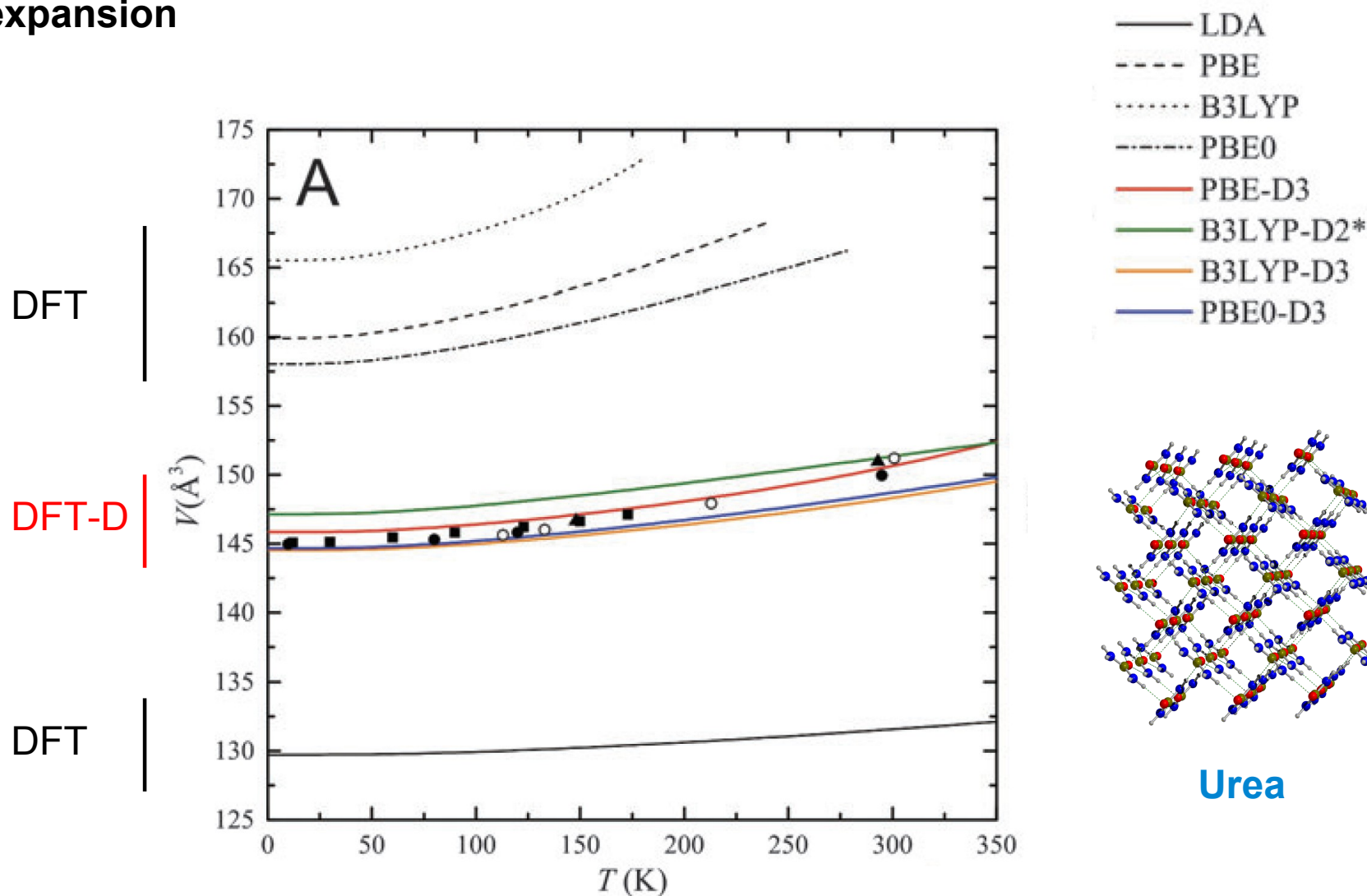
Can the QHA provide a reliable description of their thermal properties?

AE, J. Maul and B. Civalleri *Chem. Commun.*, **52**, 1820 (2016)

Quasi-Harmonic Approximation



Thermal expansion



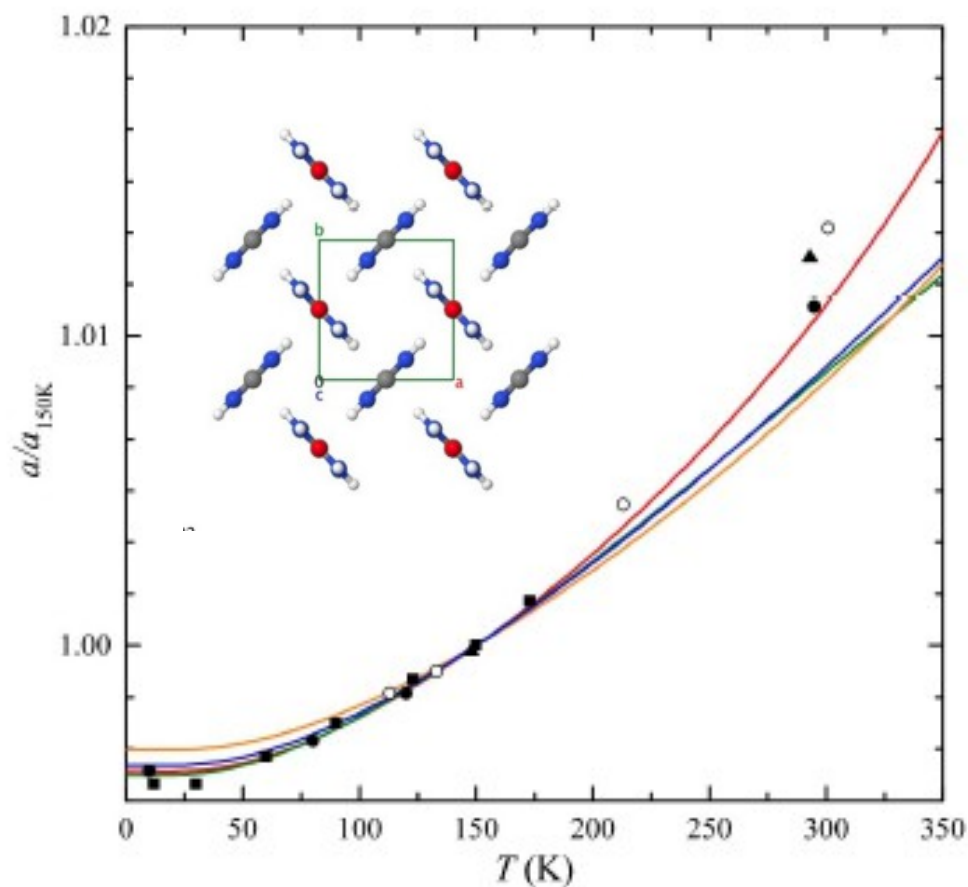
AE, J. Maul and B. Civalleri *Chem. Commun.*, **52**, 1820 (2016)

Quasi-Harmonic Approximation

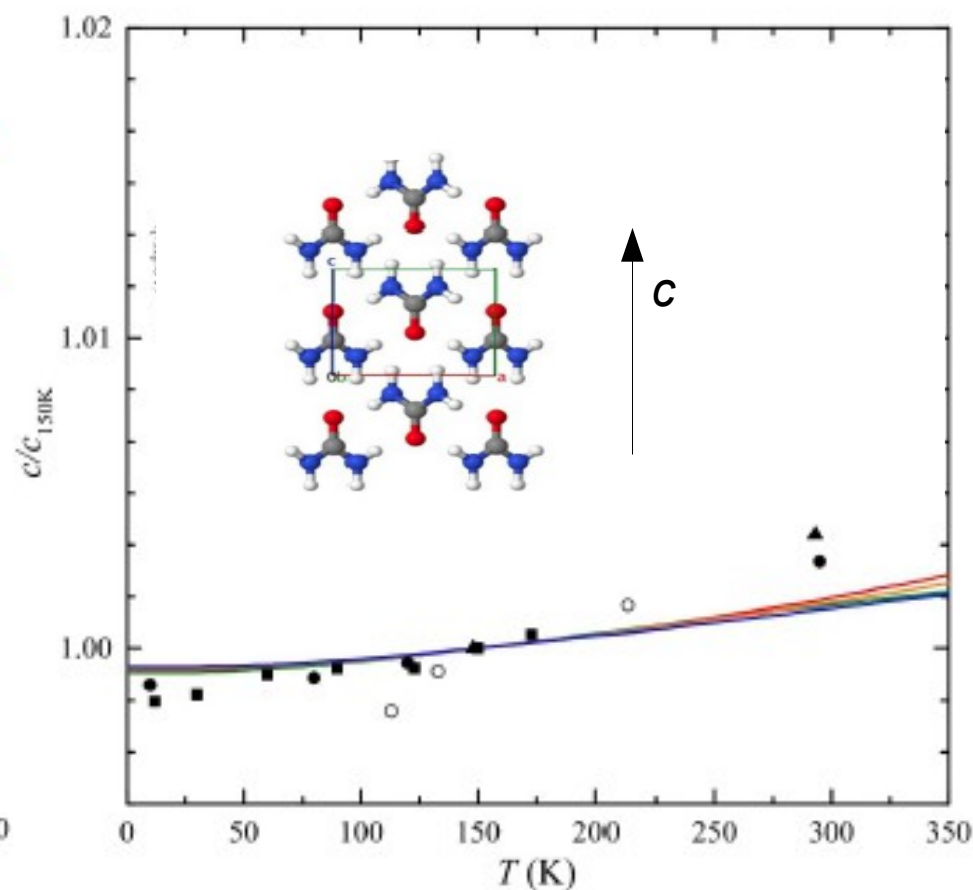


Thermal expansion (anisotropy)

a-b plane



c axis

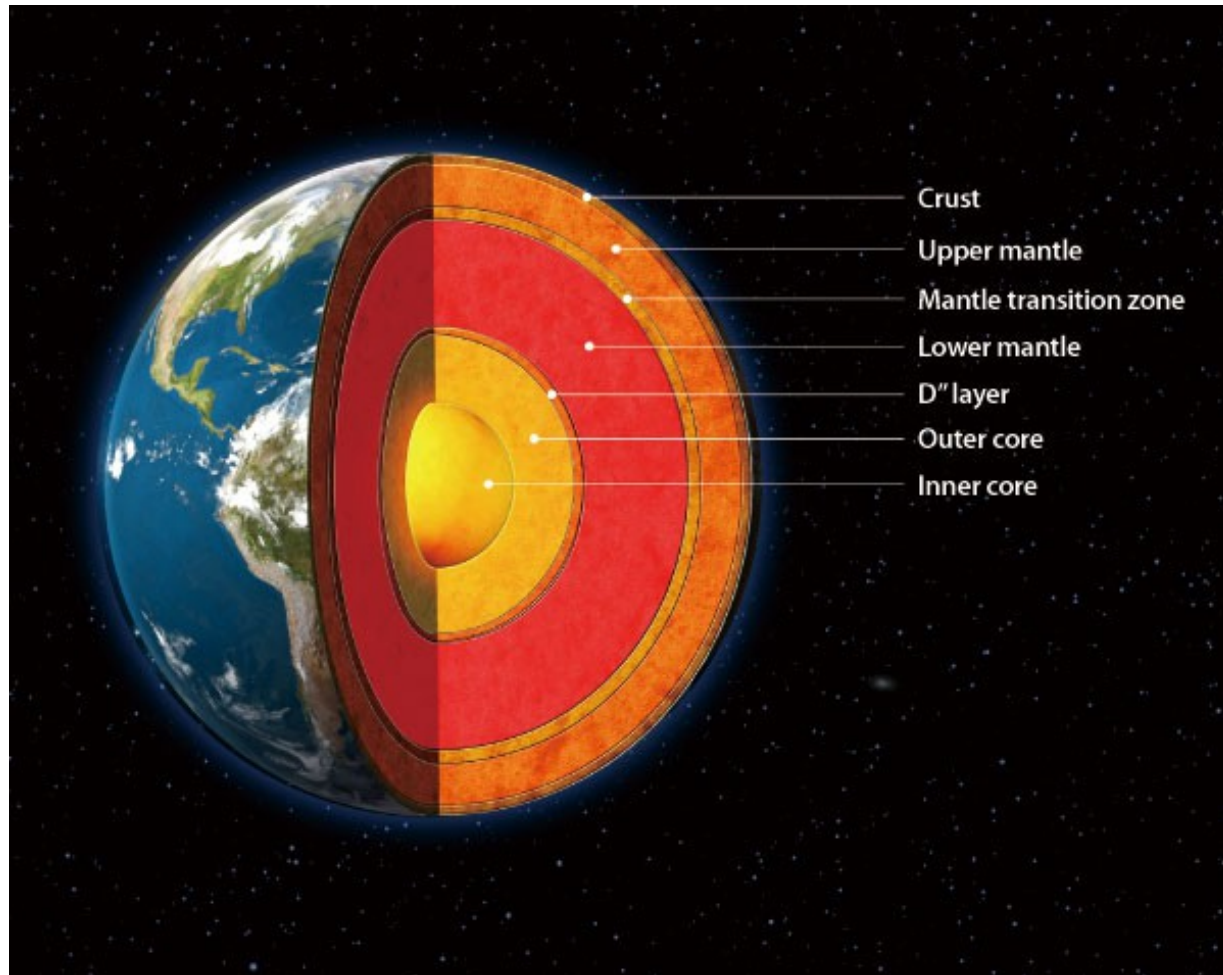


AE, J. Maul and B. Civalleri *Chem. Commun.*, **52**, 1820 (2016)

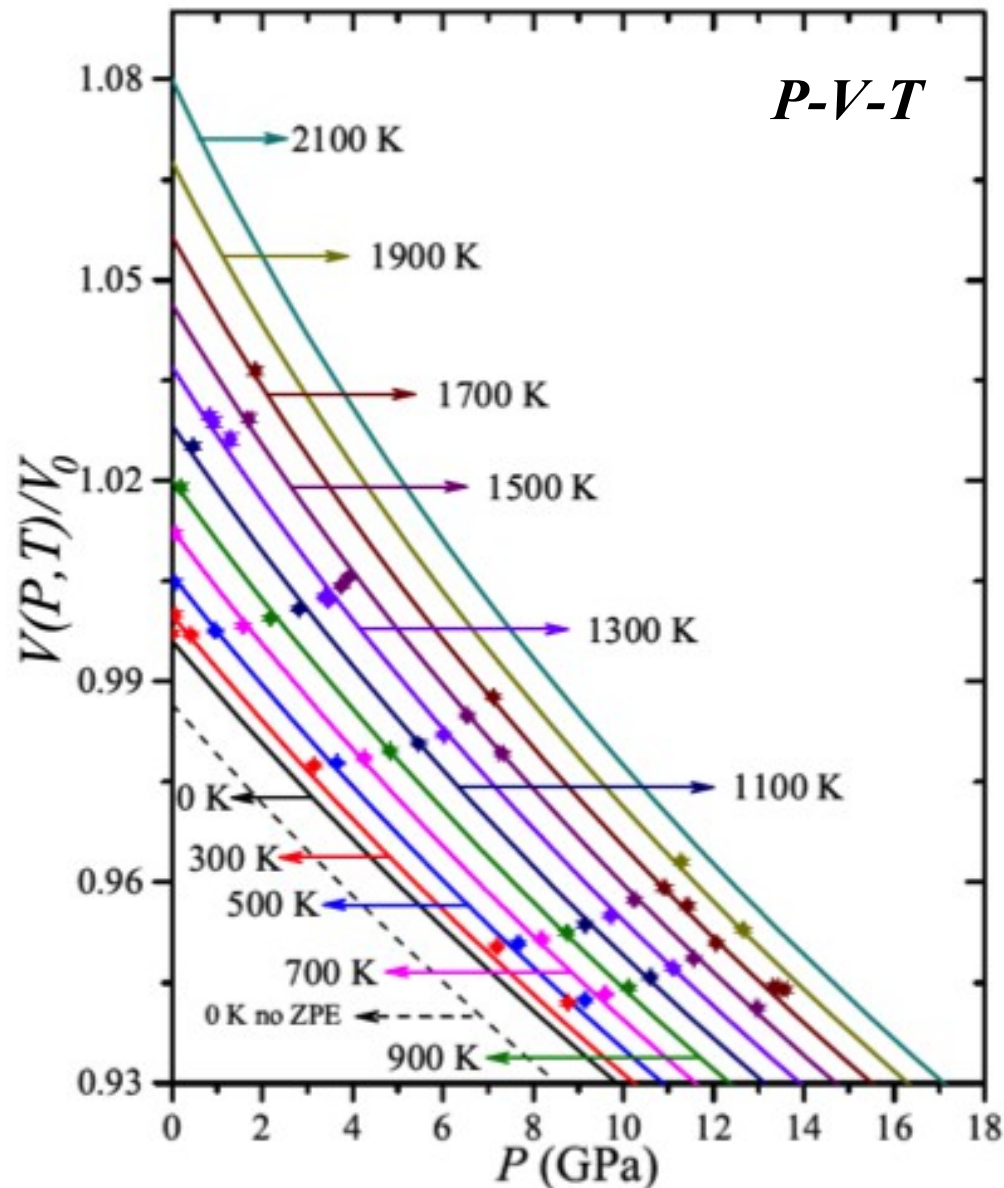
Combining Temperature and Pressure



$$P(V;T) = -\frac{\partial F^{\text{QHA}}(V;T)}{\partial V} = -\frac{\partial U_0(V)}{\partial V} - \frac{\partial F_{\text{vib}}^{\text{QHA}}(V;T)}{\partial V}$$

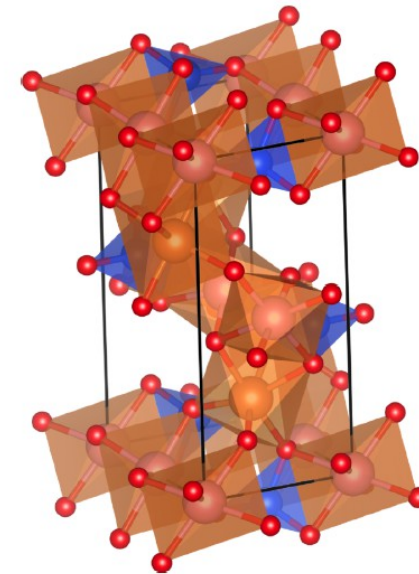


Combining Temperature and Pressure



In situ X-ray diffraction:

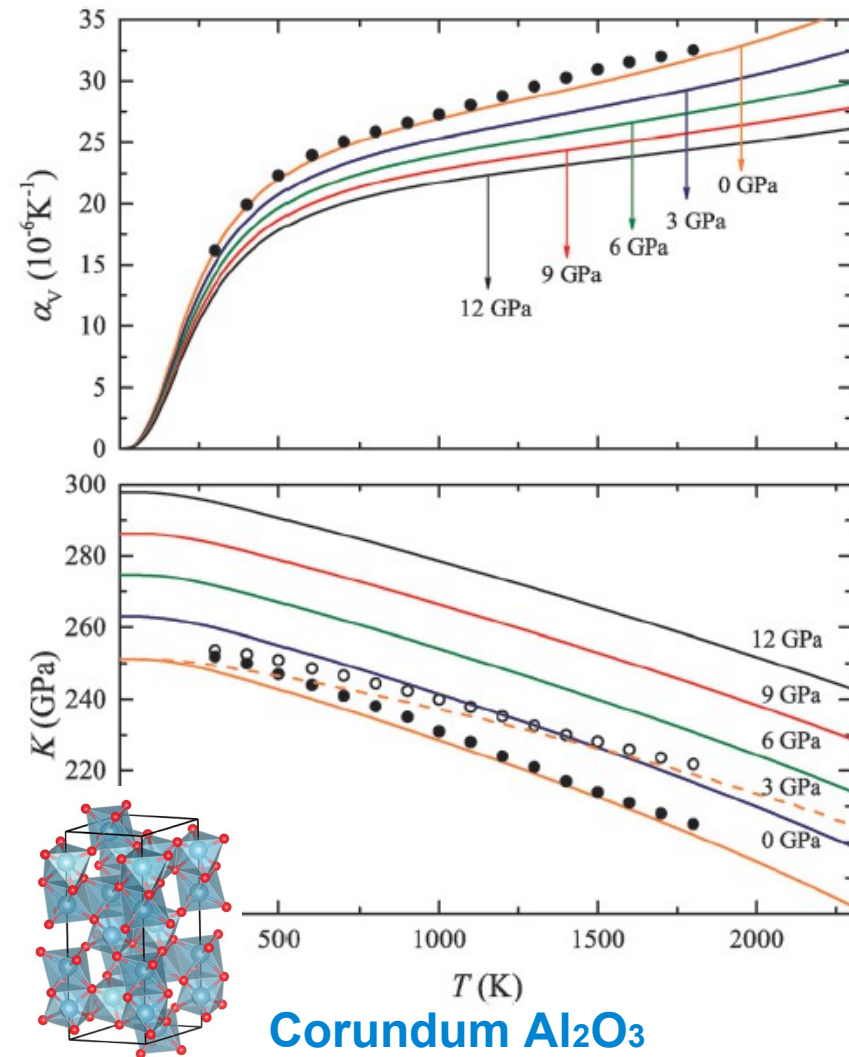
T. Katsura, A. Shatskiy, M. G. M. Manthilake, S. Zhai, H. Fukui, D. Yamazaki, T. Matsuzaki, A. Yoneda, E. Ito, A. Kuwata et al., *Phys. Earth Planet. Inter.* **174**, 86 (2009).



Forsterite Mg_2SiO_4

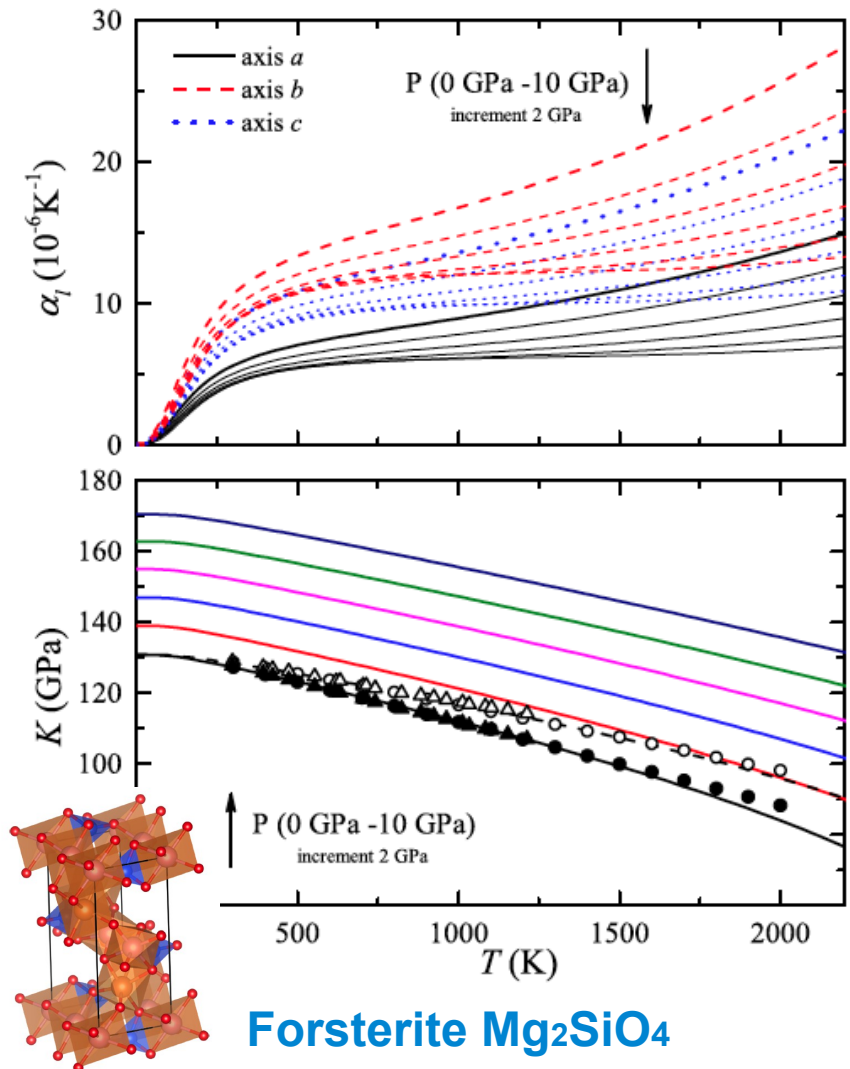
AE, J. Maul, M. De la Pierre and R. Dovesi, *J. Chem. Phys.*, **142**, 204502 (2015)

Combining Temperature and Pressure



Corundum Al_2O_3

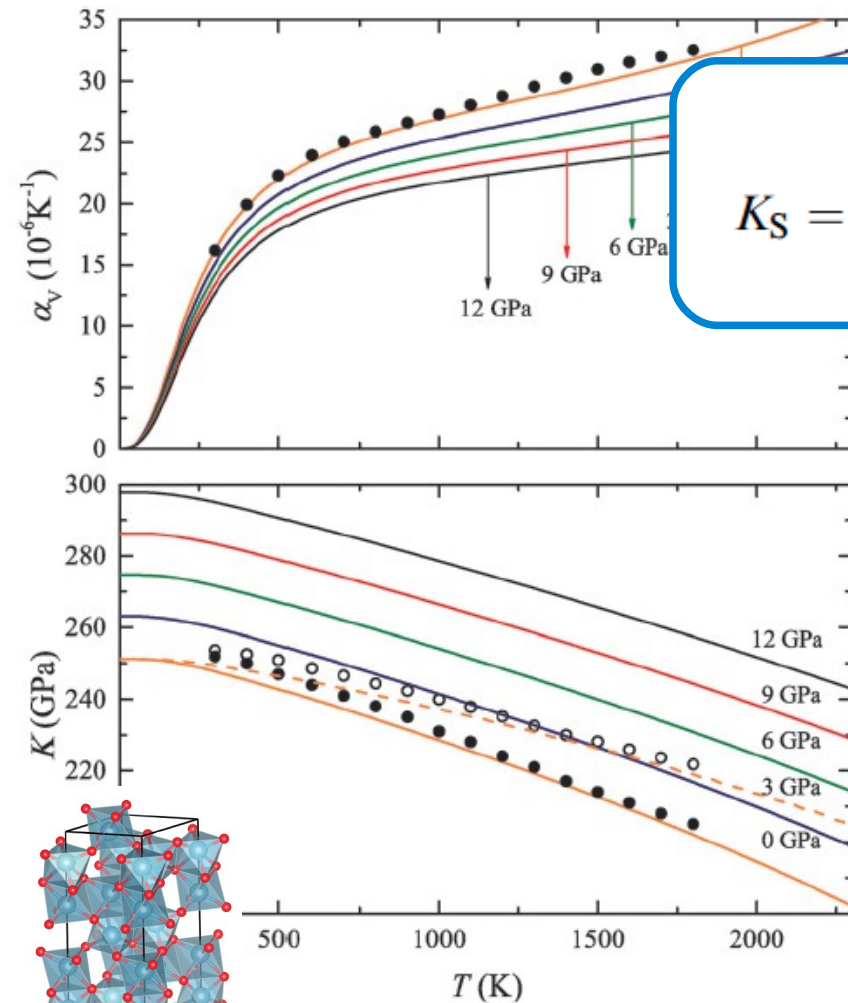
AE, J. Maul, R. Demichelis and R. Dovesi,
Phys. Chem. Chem. Phys., **17**, 11670-11677 (2015)



Forsterite Mg_2SiO_4

AE, J. Maul, M. De la Pierre and
 R. Dovesi, *J. Chem. Phys.*, **142**,
 204502 (2015)

Combining Temperature and Pressure

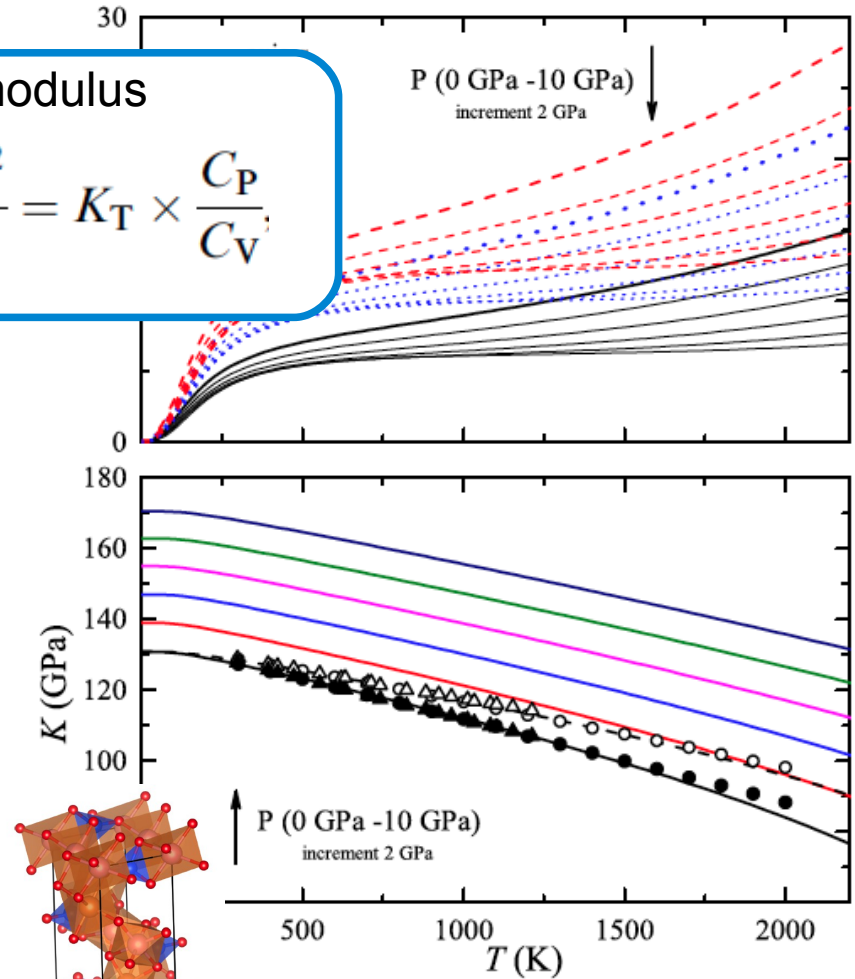


Corundum Al_2O_3

AE, J. Maul, R. Demichelis and R. Dovesi,
Phys. Chem. Chem. Phys., **17**, 11670-11677 (2015)

Adiabatic bulk modulus

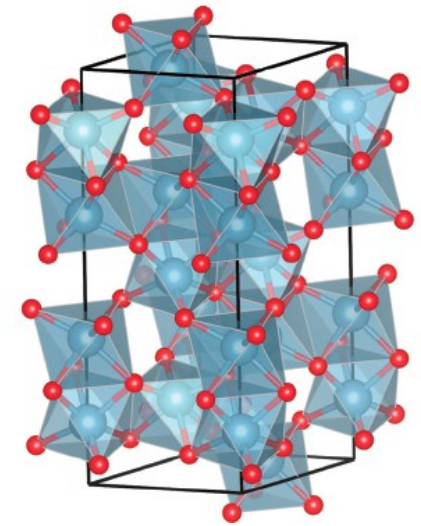
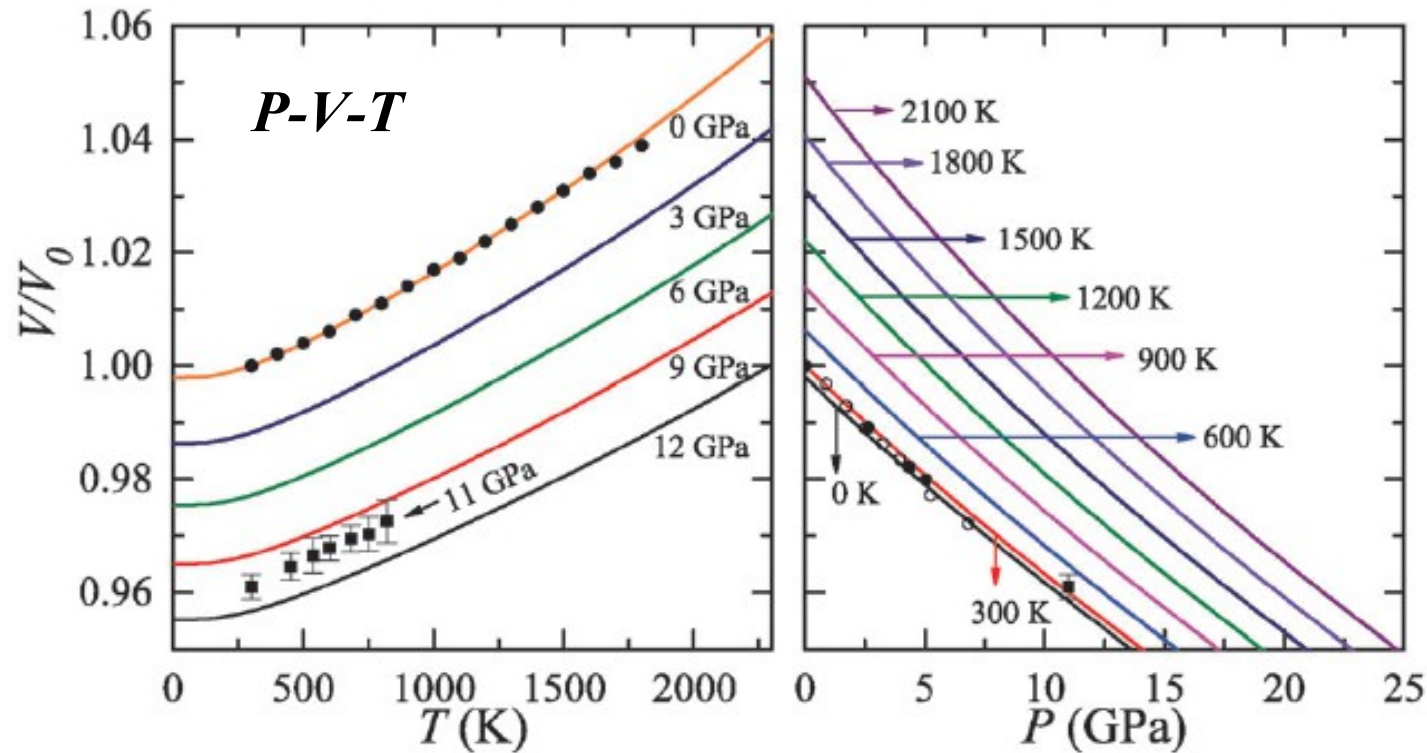
$$K_S = K_T + \frac{\alpha_V^2 V T K_T^2}{C_V} = K_T \times \frac{C_P}{C_V}$$



Forsterite Mg_2SiO_4

AE, J. Maul, M. De la Pierre and
 R. Dovesi, *J. Chem. Phys.*, **142**,
 204502 (2015)

Combining Temperature and Pressure



Corundum Al_2O_3

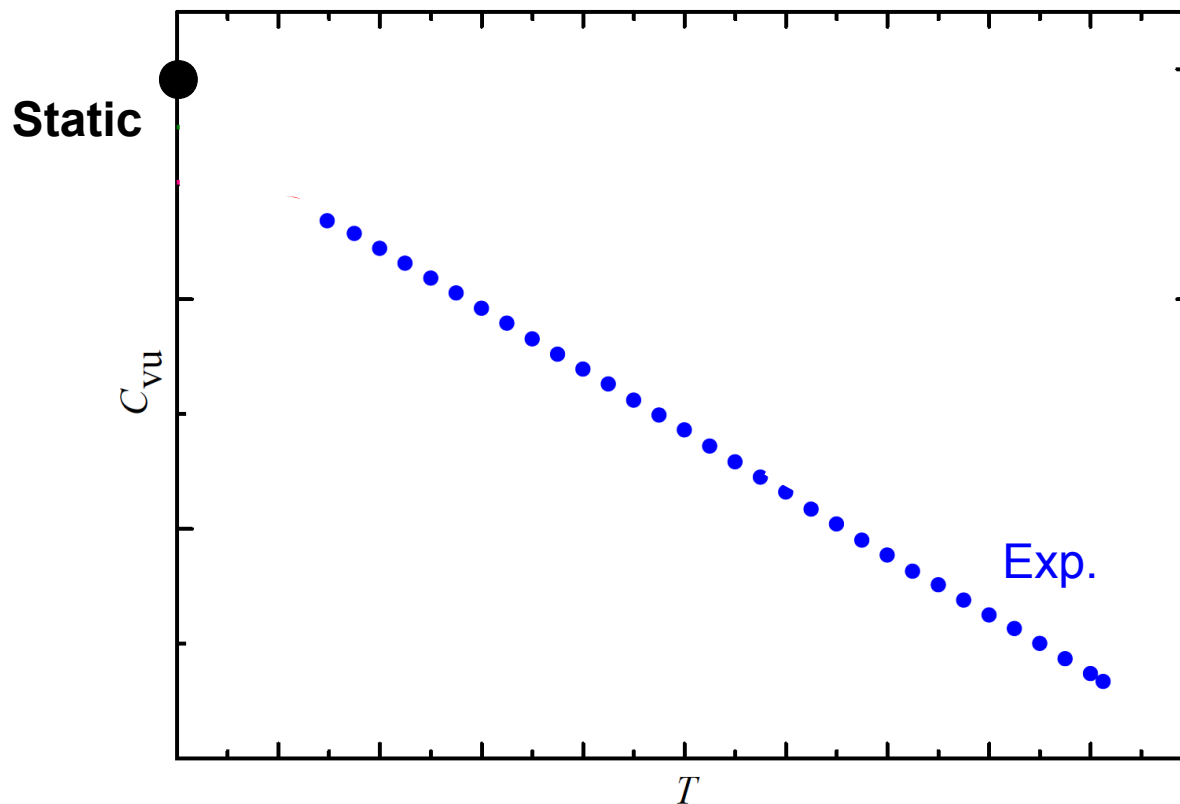
AE, J. Maul, R. Demichelis and R. Dovesi,
Phys. Chem. Chem. Phys., **17**, 11670-11677 (2015)



Mechanical properties of materials change with temperature. The elastic tensor defined as:

$$C_{vu} = \frac{1}{V_0} \left. \frac{\partial^2 E}{\partial \eta_v \partial \eta_u} \right|_{\eta=0}$$

allows to compute static, 0 K, values.

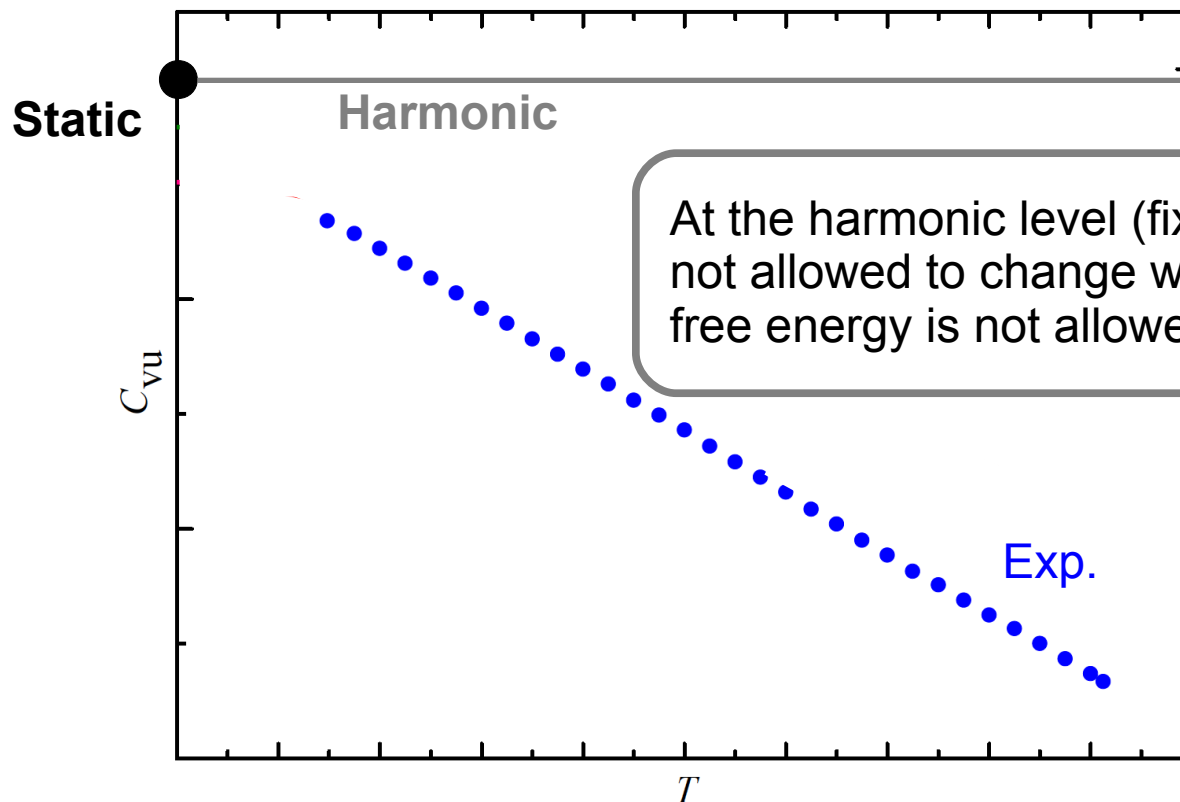




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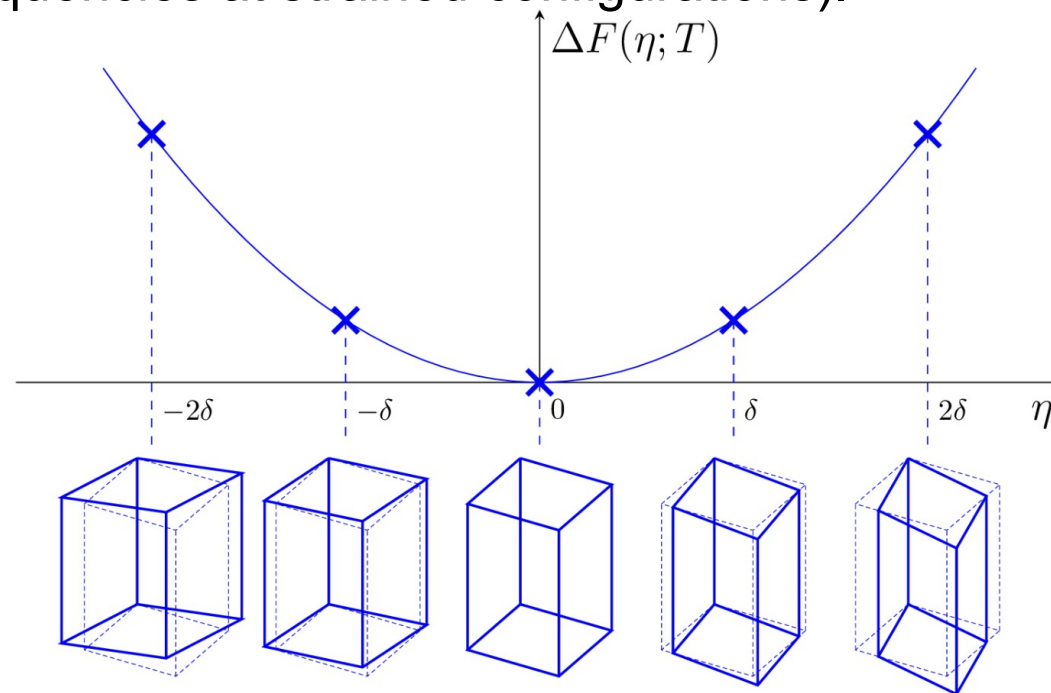




The thermal dependence of the (**isothermal**) elastic stiffness constants (and thus of all mechanical properties) of a material is defined through:

$$C_{vu}^T(T) = \frac{1}{V(T)} \left. \frac{\partial^2 F}{\partial \eta_v \partial \eta_u} \right|_{\eta=0}$$

and can be obtained from a quasi-harmonic approach. As we compute free energy derivatives with respect to strain numerically, this becomes a computationally heavy task (harmonic frequencies at strained configurations).





Adiabatic elastic constants can also be computed within a quasi-harmonic approach, that can be more directly compared to those experimentally measured by Brillouin scattering.

$$C_{vu}^S(T) = C_{vu}^T(T) + \frac{T\bar{V}(T)\lambda_v(T)\lambda_u(T)}{\bar{C}_V(T)}$$

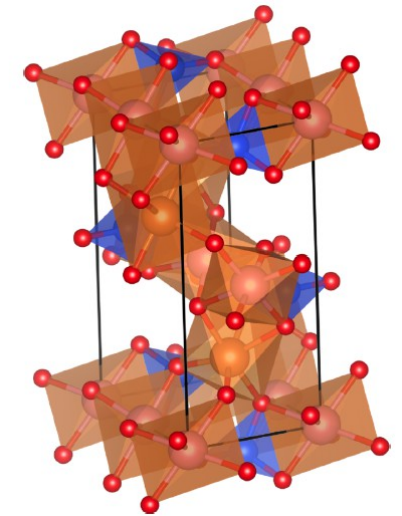
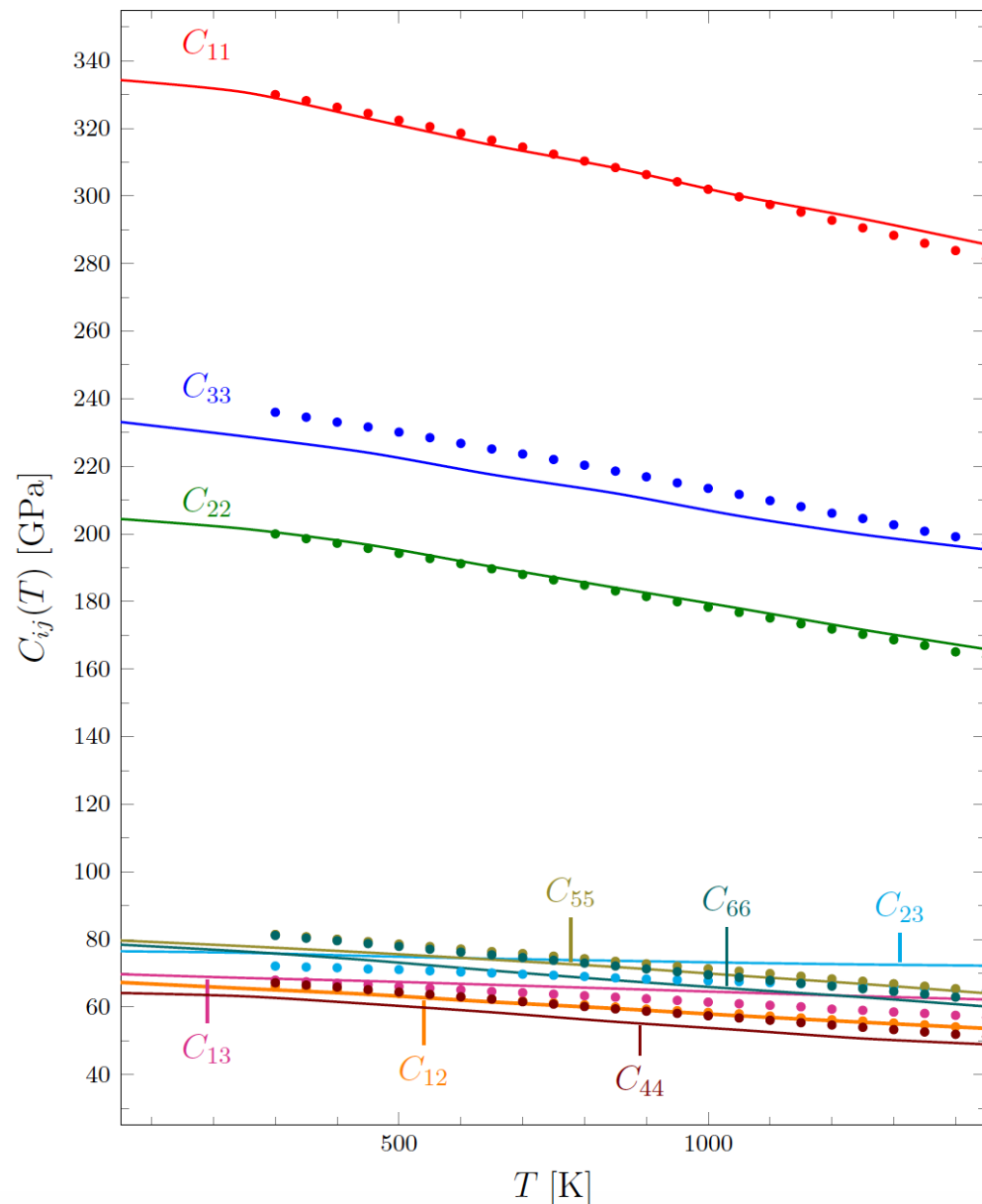
with

$$\lambda_v(T) = \left. \frac{\partial S}{\partial \eta_v} \right|_T = - \left. \frac{\partial \sigma_v}{\partial T} \right|_\eta = - \sum_u \left. \frac{\partial \eta_u}{\partial T} \right|_\sigma \left. \frac{\partial \sigma_v}{\partial \eta_u} \right|_T = - \sum_u \alpha_u(T) C_{vu}^T(T)$$

An algorithm has been implemented in **CRYSTAL23** for the quasi-harmonic evaluation of isothermal and adiabatic elastic constants.

M. Destefanis, C. Ravoux, A. Cossard, AE
Minerals., **9**, 16 (2019)

Thermo-Elasticity



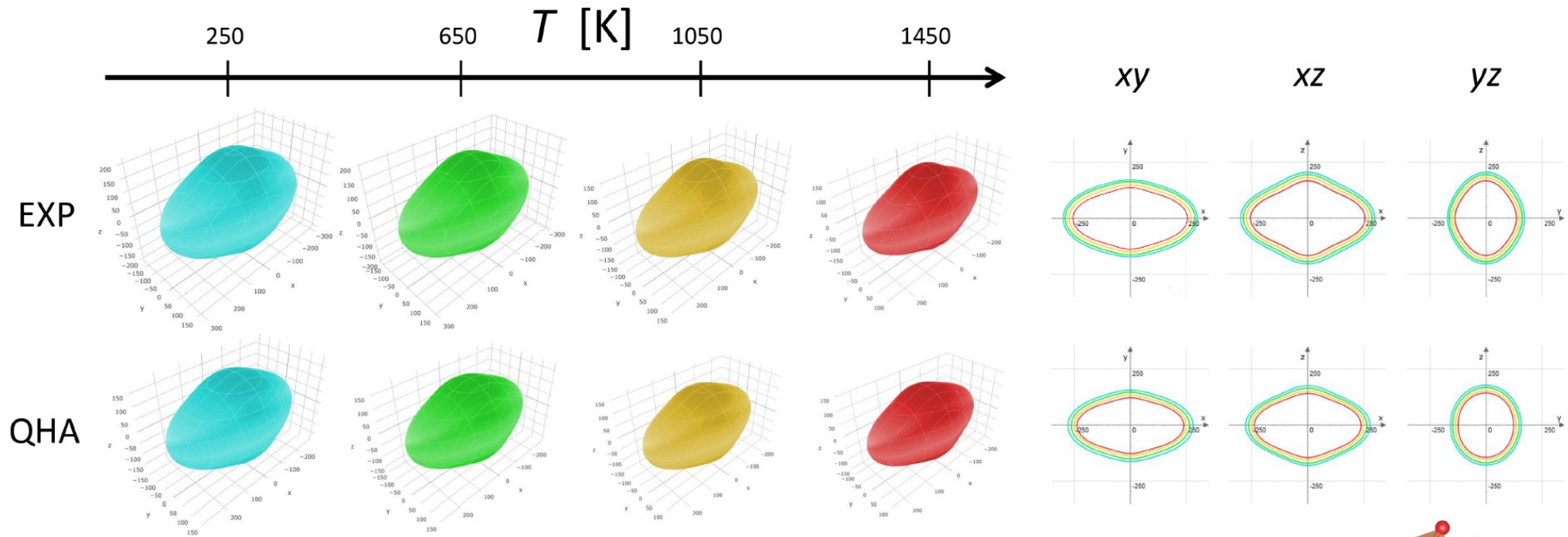
Forsterite Mg_2SiO_4

Adiabatic elastic constants of Forsterite as a function of temperature as computed with the quasi-harmonic approximation and compared to experimental values.

B3LYP functional.

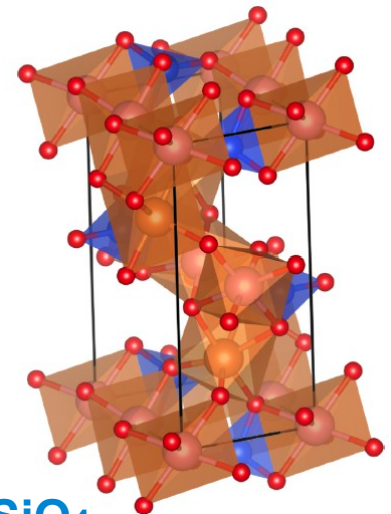
M. Destefanis, C. Ravoux, A. Cossard, AE *Minerals.*, **9**, 16 (2019)

Thermo-Elasticity



3D and 2D view of the effect of temperature on the Young modulus of forsterite.

M. Destefanis, C. Ravoux, A. Cossard, AE
Minerals., **9**, 16 (2019)



Forsterite Mg_2SiO_4



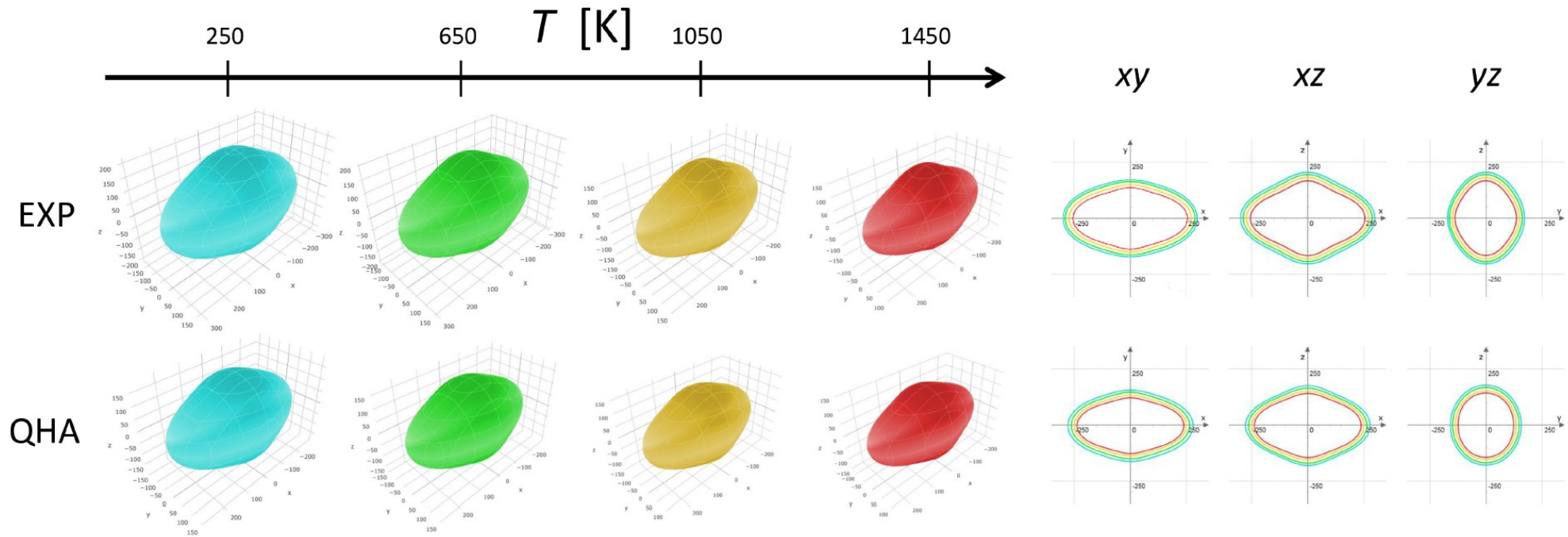
A computationally **(much!) cheaper** approximation can be introduced. Instead of computing the isothermal elastic constants from a **full quasi-harmonic approach** as:

$$C_{vu}^T(T) = \frac{1}{V(T)} \left. \frac{\partial^2 F}{\partial \eta_v \partial \eta_u} \right|_{\eta=0}$$

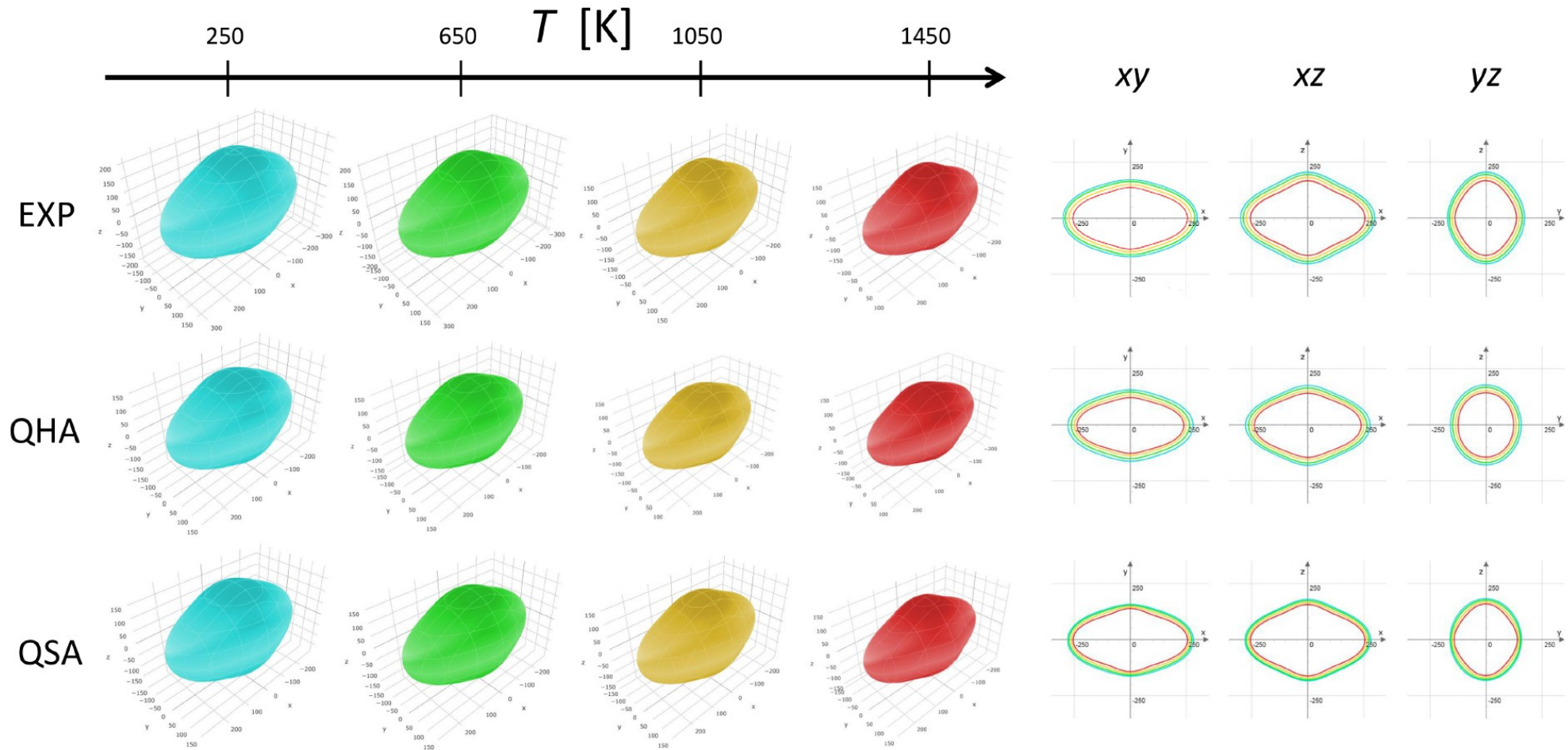
We can use a **quasi-static approximation** where thermal expansion is accounted for while neglecting thermal contributions to the free energy derivatives with strain:

$$C_{vu}^T(T) = \frac{1}{V(T)} \left. \frac{\partial^2 E}{\partial \eta_v \partial \eta_u} \right|_{\eta=0}$$

Thermo-Elasticity



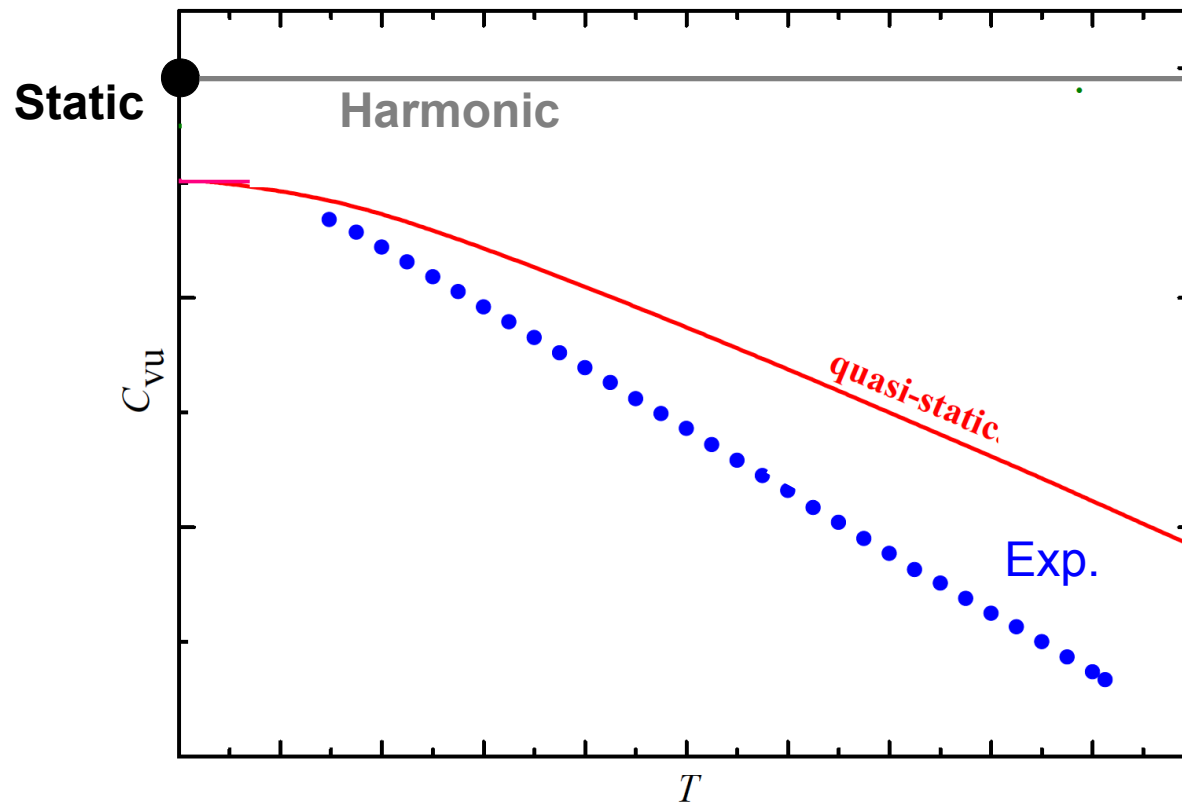
Thermo-Elasticity



Thermo-Elasticity



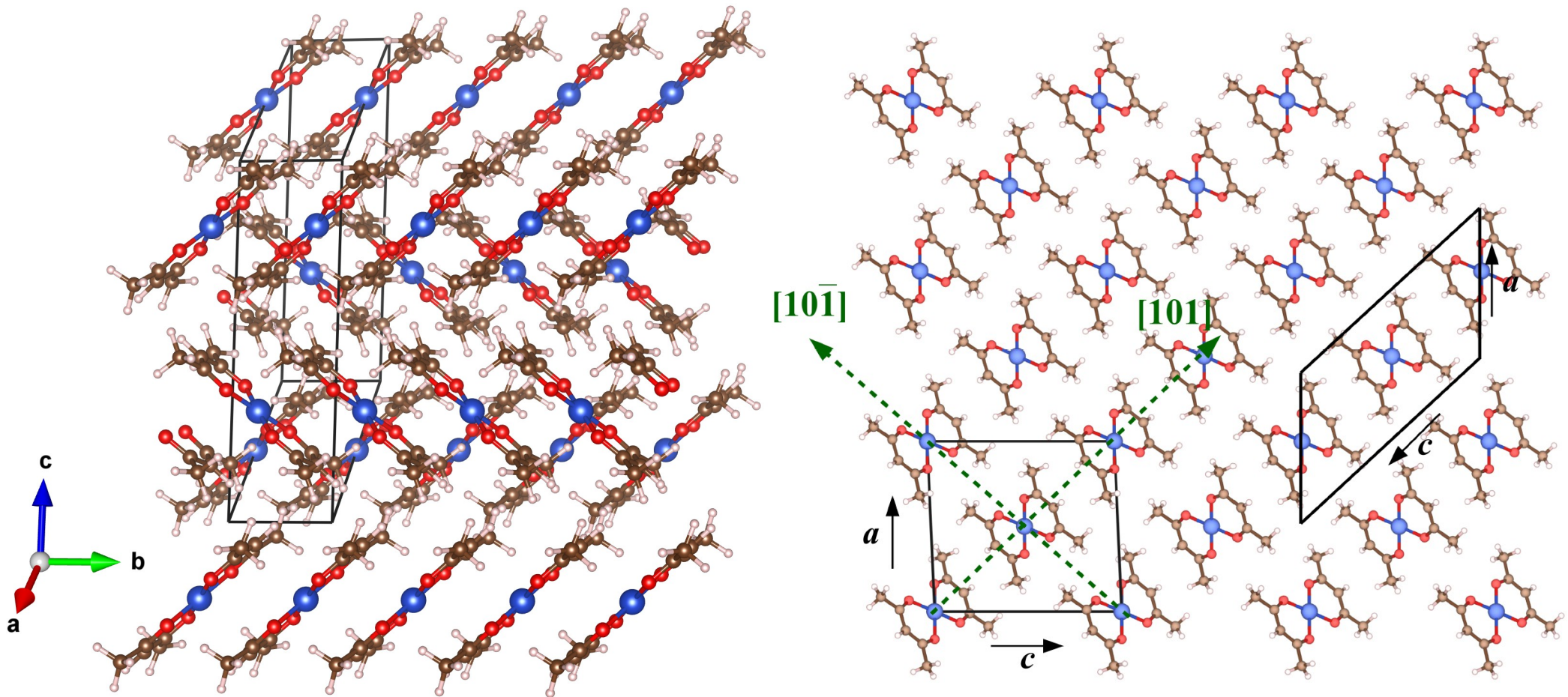
The QSA is able to describe a large fraction of thermal effects.



Thermo-Elasticity



Copper Acetylacetonate Crystals





Copper Acetylacetonate Crystals

Young Modulus

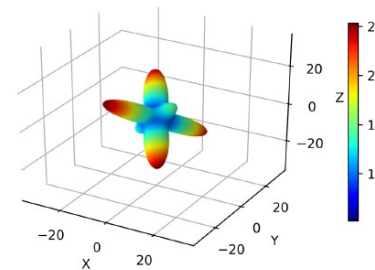
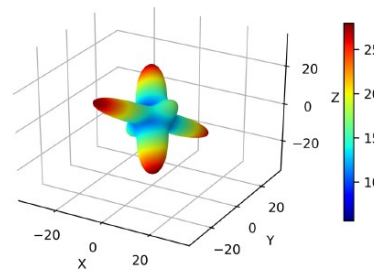
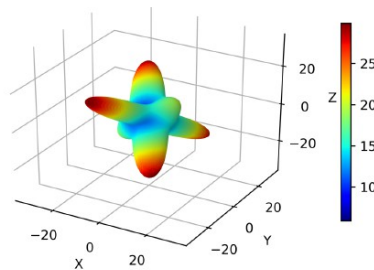
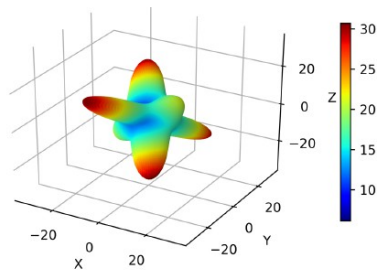
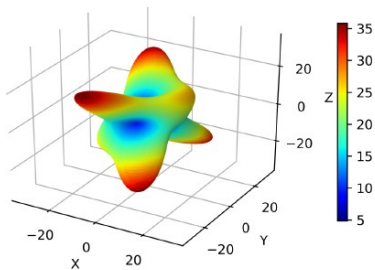
0 K no ZPE

0 K

100 K

200 K

300 K



T (in K)	$E_{[10\bar{1}]}$ (in GPa)	$E_{[101]}$ (in GPa)
0 static	25.3	5.3
0 with ZPE	18.6	7.1
100	16.7	6.8
200	13.8	6.6
300	11.3	6.2
300 (Exp.)	11.3 - 13.8	4.8 - 6.9

J. Maul, D. Ongari, S. M. Moosavi, B. Smit, AE
J. Phys. Chem. Lett., **11**, 8543--8548 (2020)



Thank You for your Attention